A 9.3nV/rtHz 20b 40MS/s 94.2dB DR Signal-Chain Friendly Precision SAR Converter

Rares Bodnar, Henry Kennedy, Christopher Peter Hurrell, Asif Ahmad, Mark Vickery, Luke Smithers, William Buckley, Monsoon Dutt, Pasquale Delizia, Derek Hummerston, Pawel Czapor



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Outline

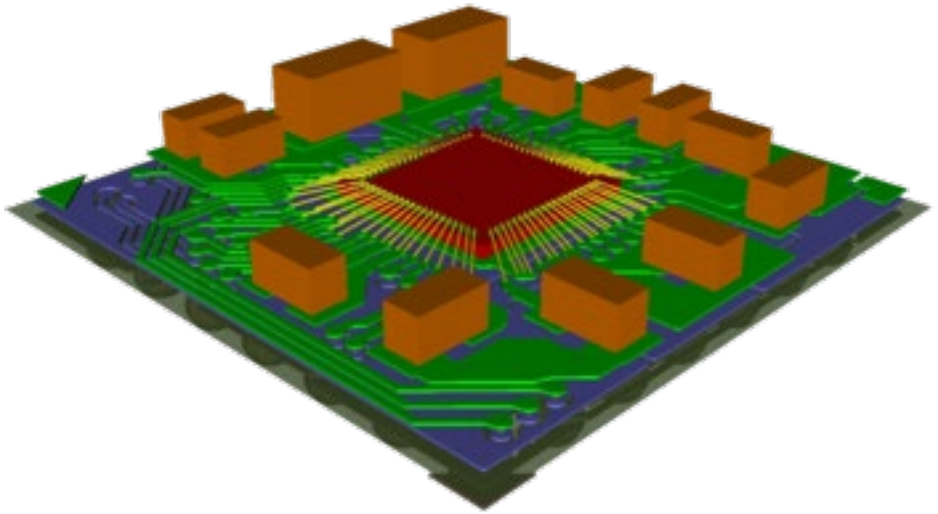
- **Motivation**
- **Power efficient, low noise architecture**
  - Precision vs. high-speed tradeoffs
  - Architecture overview
- **Precision linearity at >10Msps**
  - Reference buffer
  - Multi-dimensional dynamic element matching
- **Measurements results**
- **Conclusion**

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Project Overview

Precision, Low noise and Efficiency

- Genuine 20b LF characteristics
- Efficient, low noise ADC
- Signal range DC – 5MHz (Nyquist capable)
- Ultra-low 1/f noise

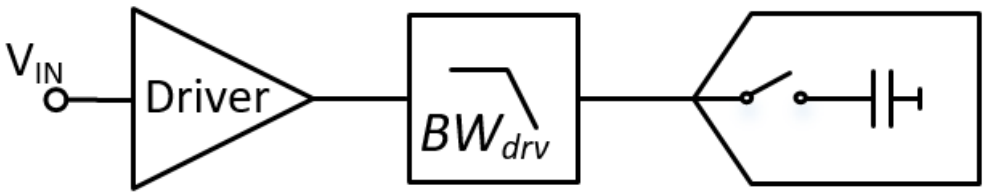


Signal chain focus

- High signal chain level FOM
- Easy to drive analog and reference inputs

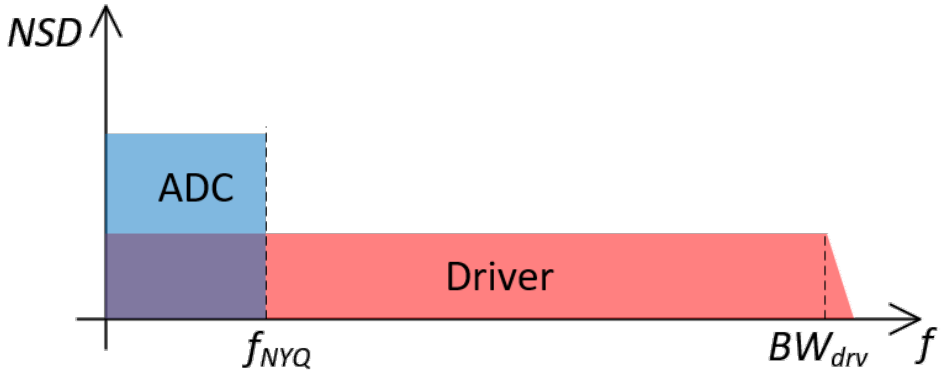
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Signal Chain SNR Trade-Offs



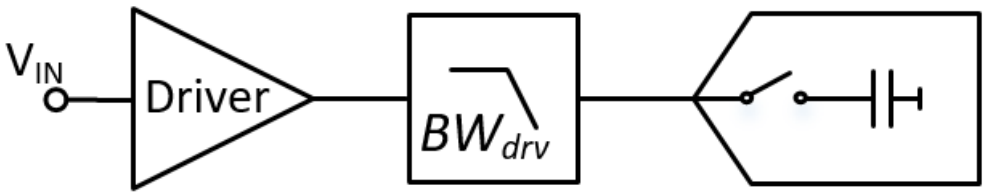
$$SNR = 20log_{10}\left(\frac{V_{PP}/\sqrt{8}}{\sqrt{NSD_{ADC}^2 f_{NYQ} + NSD_{DRV}^2 \boldsymbol{BW_{DRV}}}}\right)$$





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Signal Chain SNR Trade-Offs



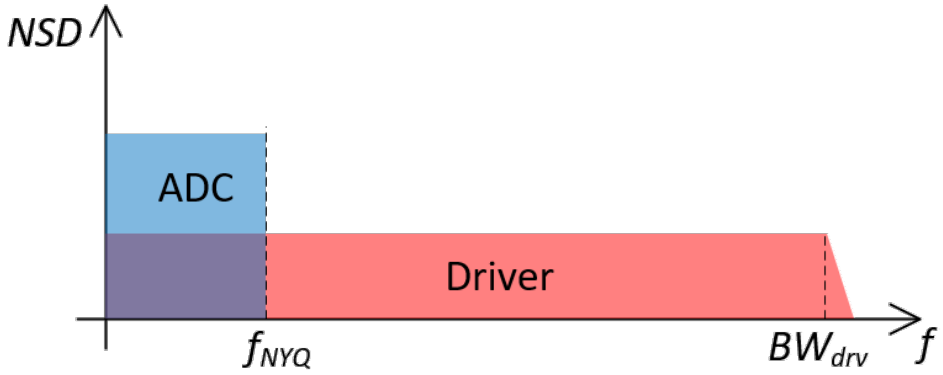


$$SNR = 20log_{10}\left(\frac{V_{PP}/\sqrt{8}}{\sqrt{NSD_{ADC}^2 f_{NYQ} + NSD_{DRV}^2 \boldsymbol{BW_{DRV}}}}\right)$$

20b settling require $\sim 14\tau \rightarrow BW_{DRV} = \frac{14}{2\pi T_{ACQ}}$

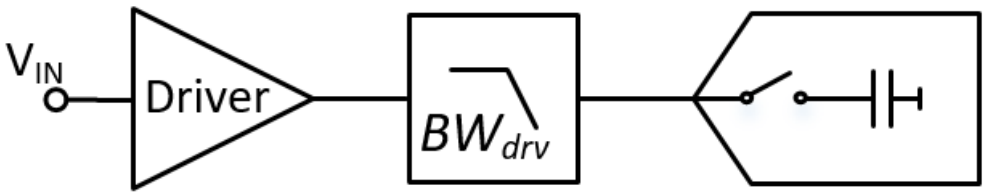
For $T_{ACQ} = \frac{1}{4f_{CLK}} \rightarrow$

$$\boldsymbol{ENBW_{DRV} = 28 f_{NYQ}}$$

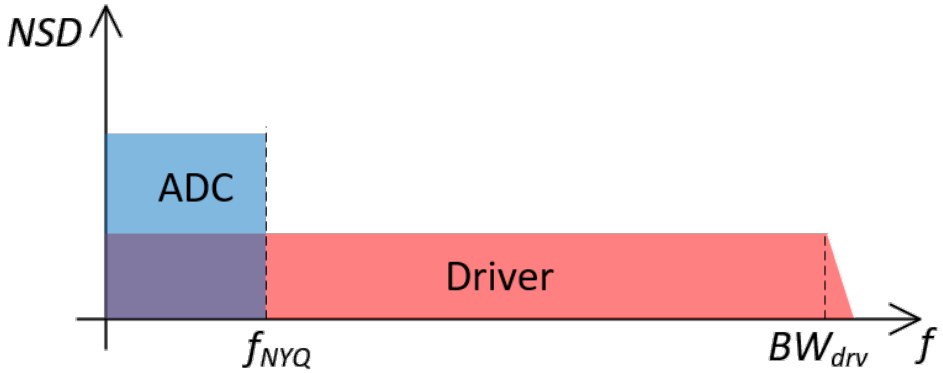


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Post aliasing:

$$SNR = 20log_{10}\left(\frac{V_{PP}/\sqrt{8}}{\sqrt{f_{NYQ}\left(NSD_{ADC}^2 + \mathbf{5.3^2} NSD_{DRV}^2\right)}}\right)$$

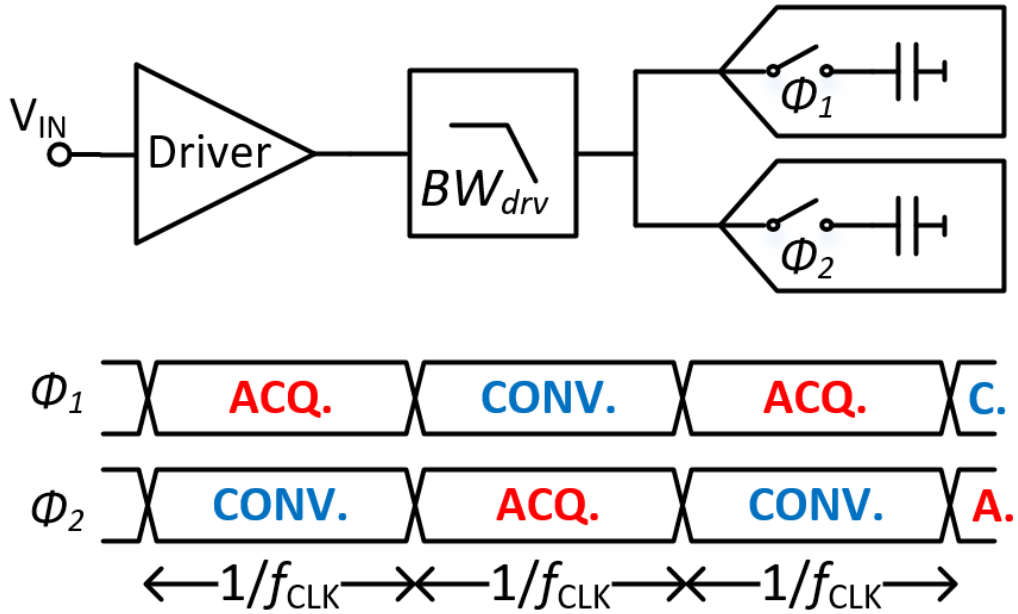
Driver must be wide BW and very low noise

→ Power hungry → Low Sig. Ch. FOM

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Mitigating the Driver Challenges

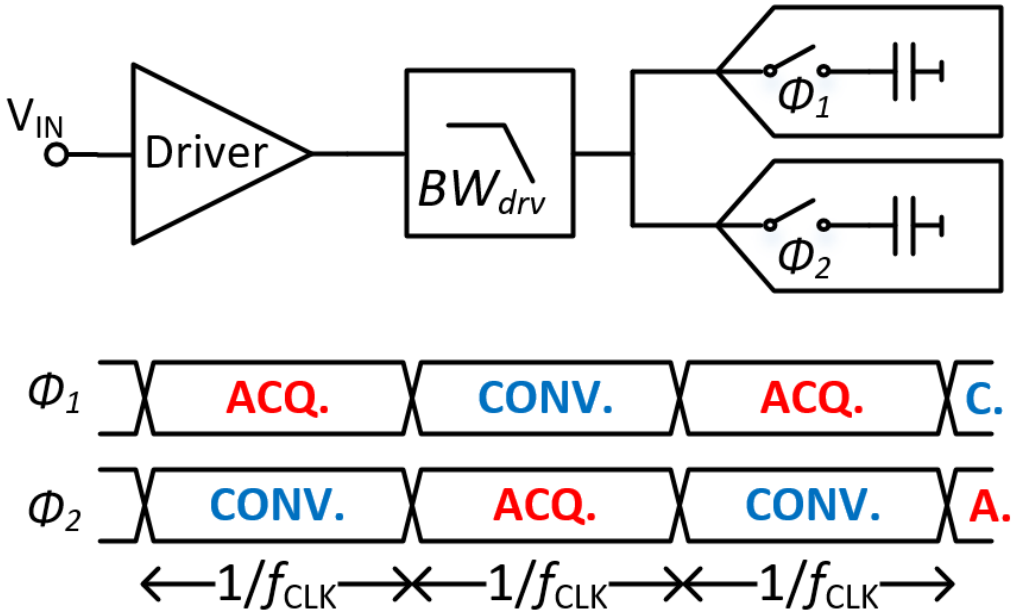
- **Time interleaved acquisition**
  - Extends the ACQ time by ~4x
    - Conventionally T_{ACQ}=25% T_{CLK}



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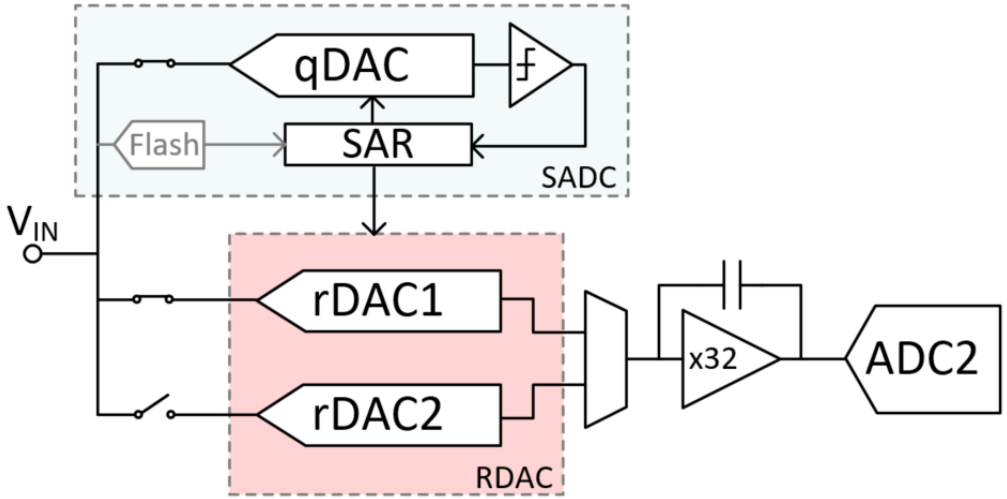
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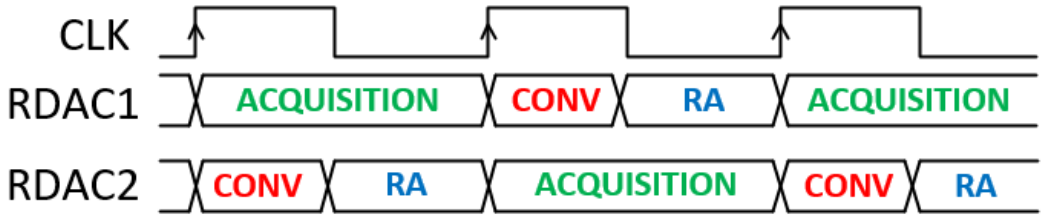
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ADC Architecture

SAR split structure

- ±3V input range → 3V first stage (0.4µm)
- **SAR quantizer** (SADC) – small and fast
- **Residue DAC** (RDAC) – SADC follower
  - Low kT/C and DEM delay
- Matched slice DAC structure
  - >12b AC matching (SADC to RDAC)





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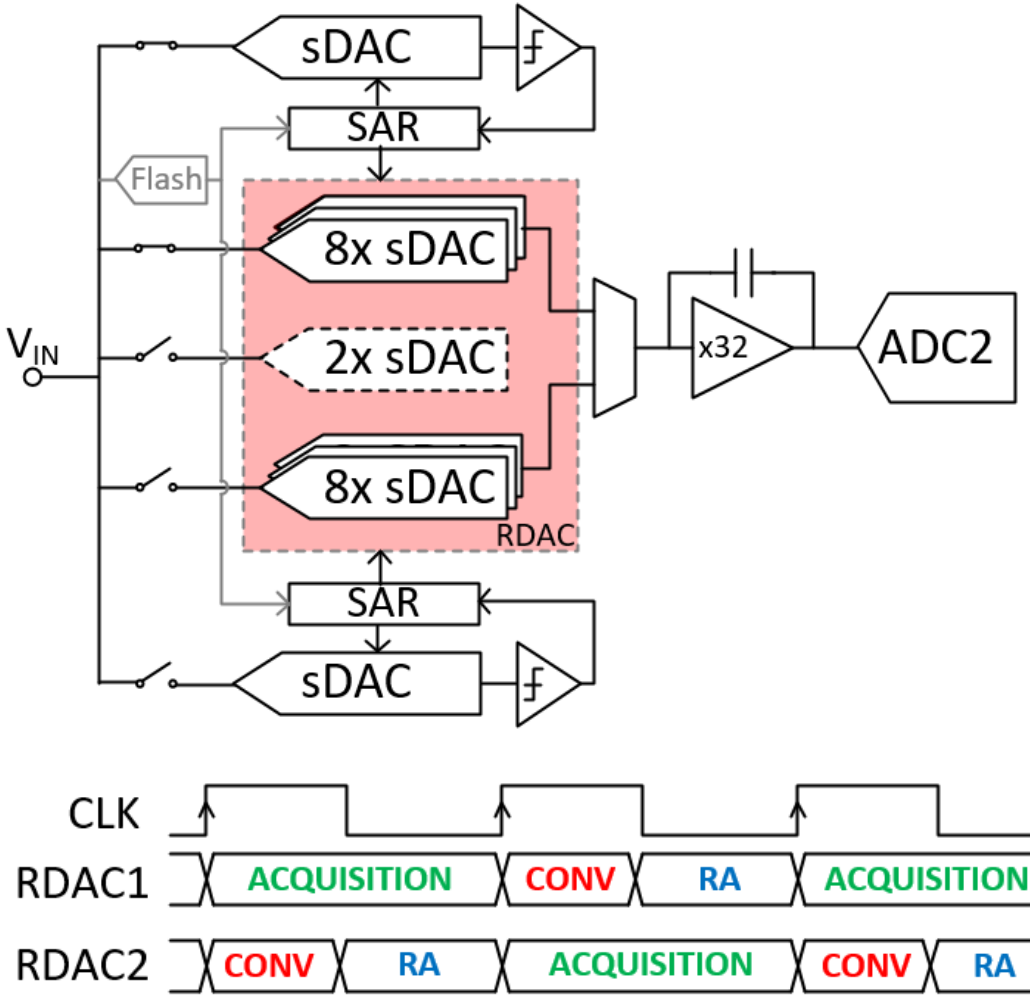
Analog Domain Interleaving (ADI)

Slice DAC structure

Quantizer (1x) and RDAC (8x)

Pool of 18x slice DACs for residue DAC

2x spare (randomisation) DACs



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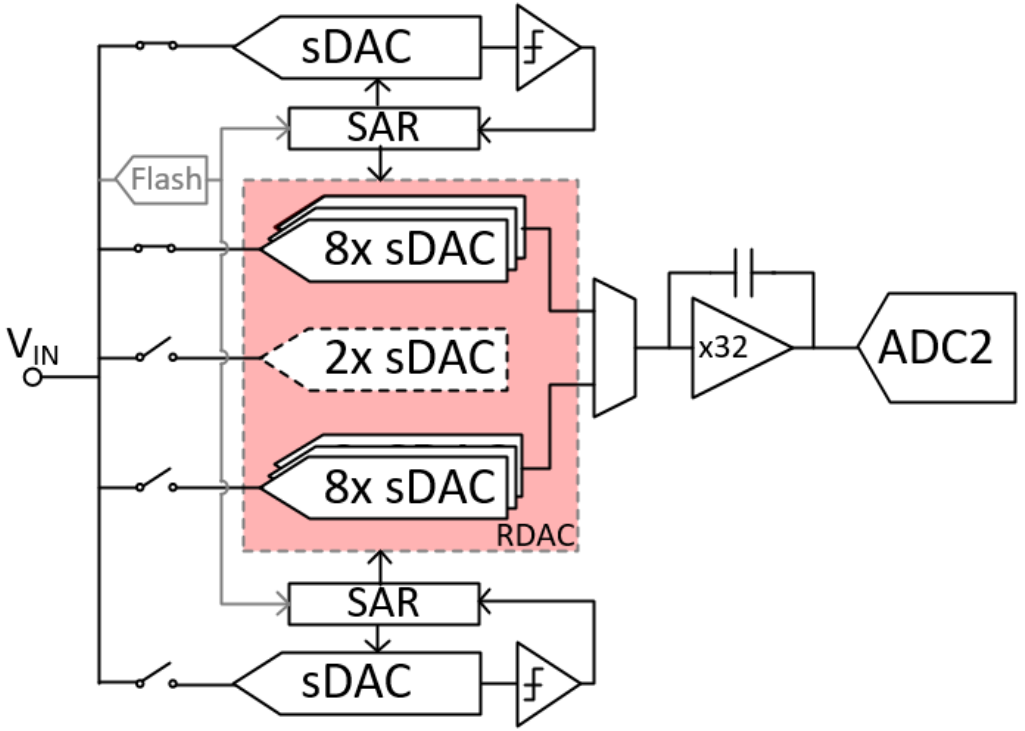
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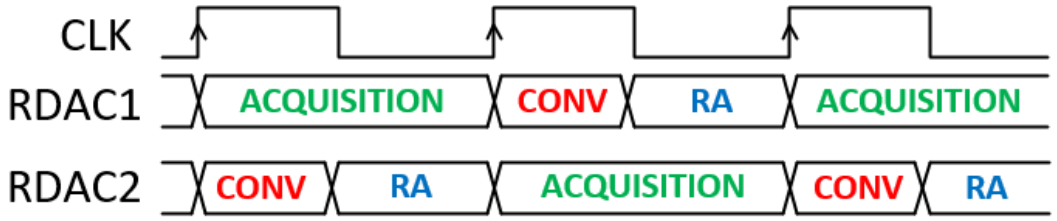
Slice DAC structure

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Shared residue amplifier

- 40% of ADC power consumption
- Matched offset and gain error
- Shared ADC2





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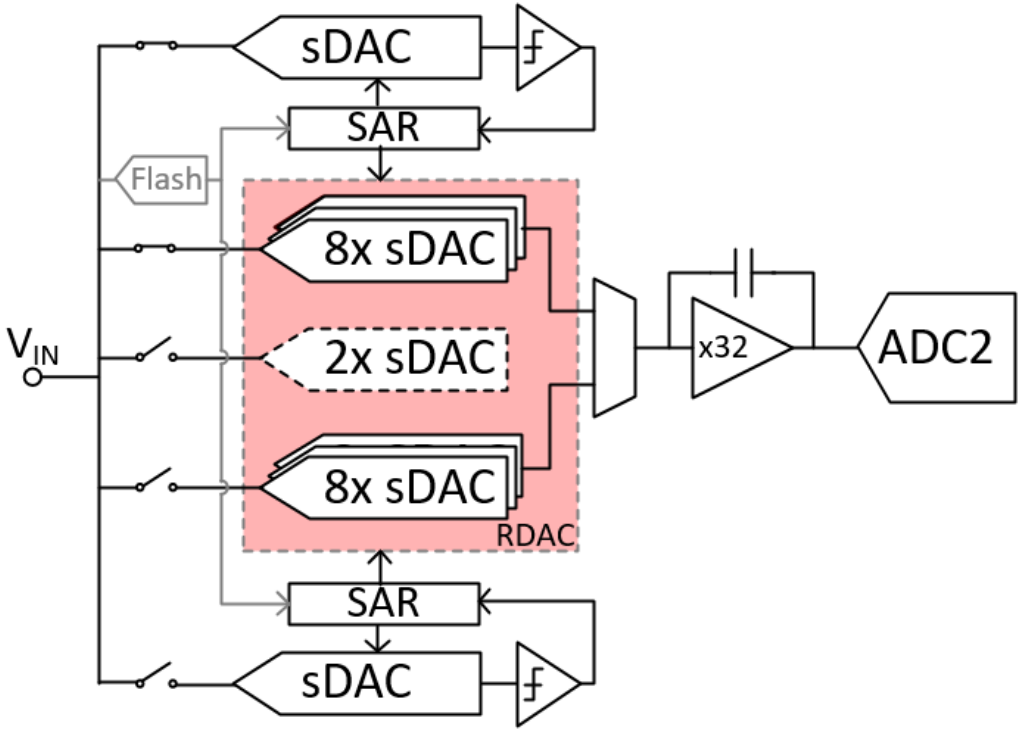
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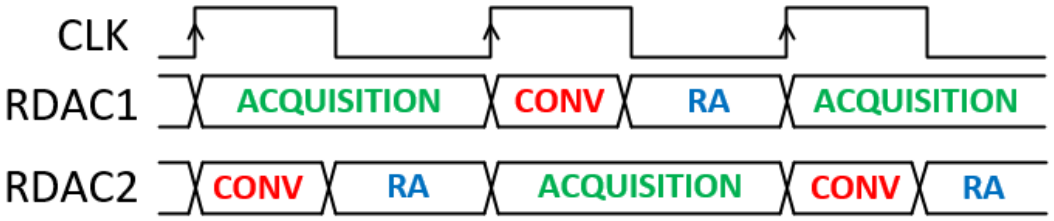
Matched offset and gain error

Shared ADC2

Second quantizer

Removes the mid-cycle input disturbance



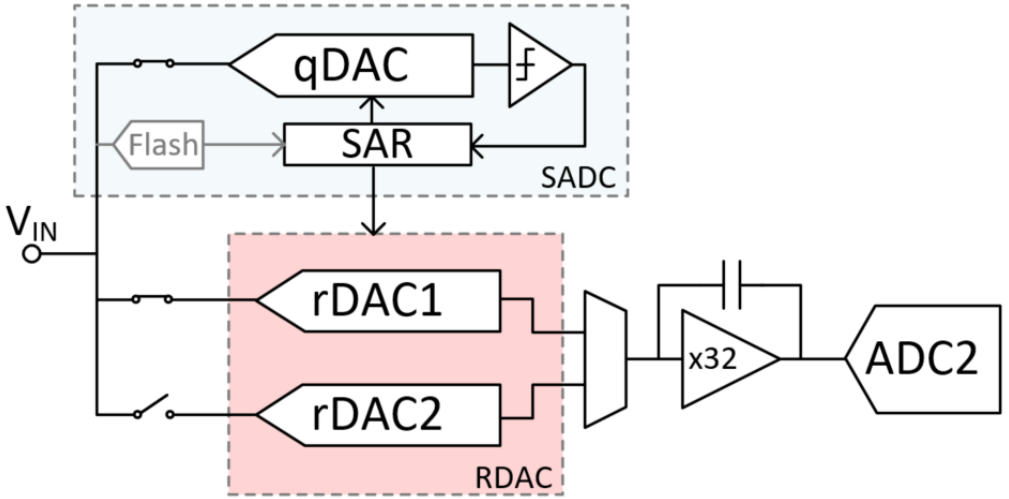


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RDAC Architecture Benefits

Minimal reference current

RDAC set directly to final state



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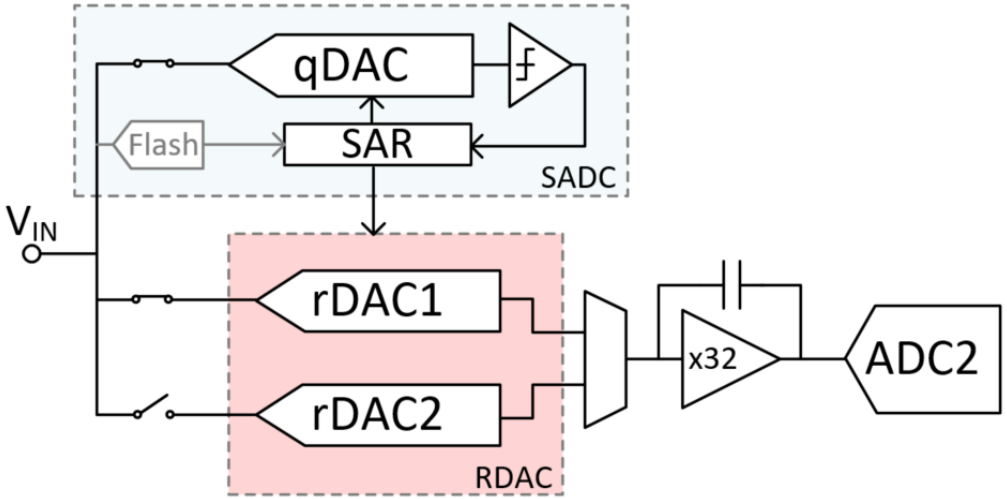
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Smaller RDAC reference switches

Improved distortion



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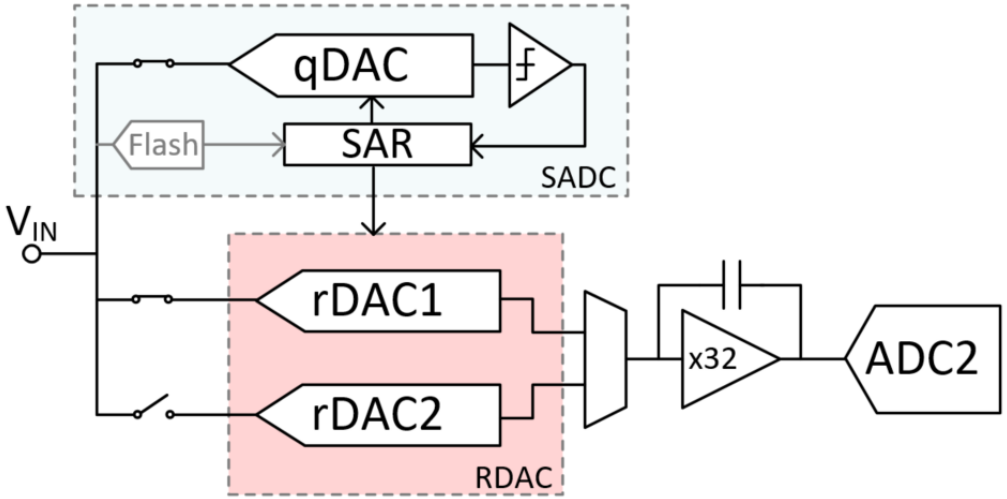
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Soft reference switching

Reduced ringing and noise



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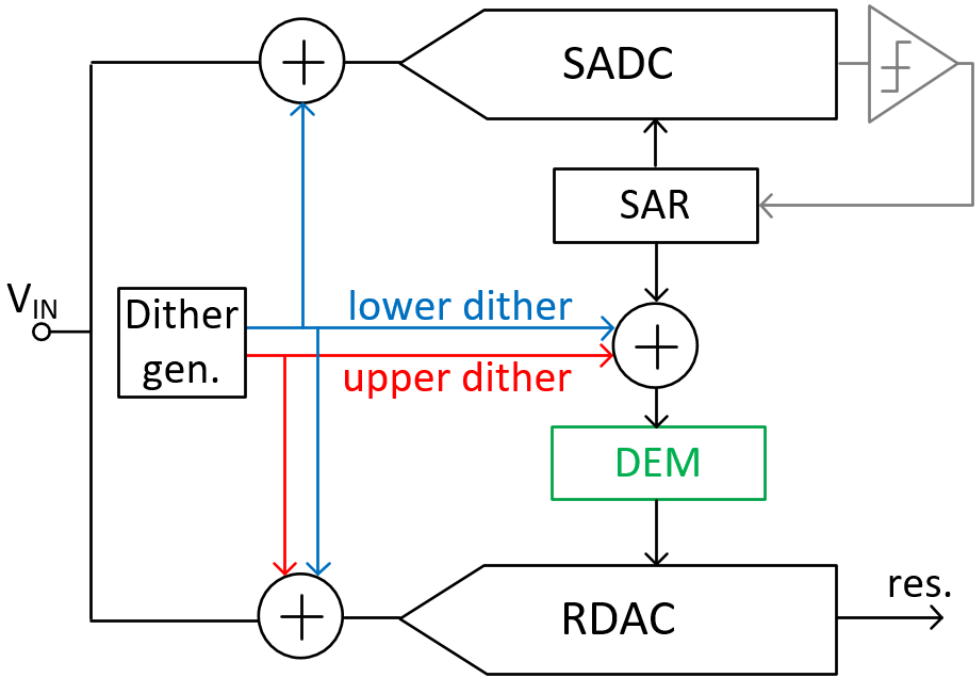
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RDAC segmentation and DEM



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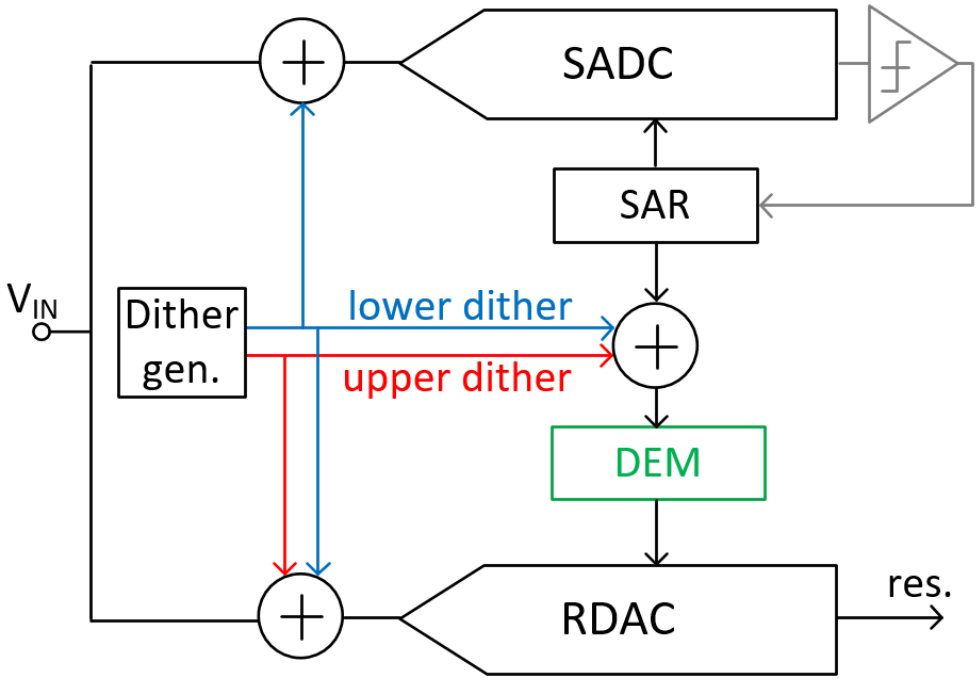
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Extra dither range

Improved linearity



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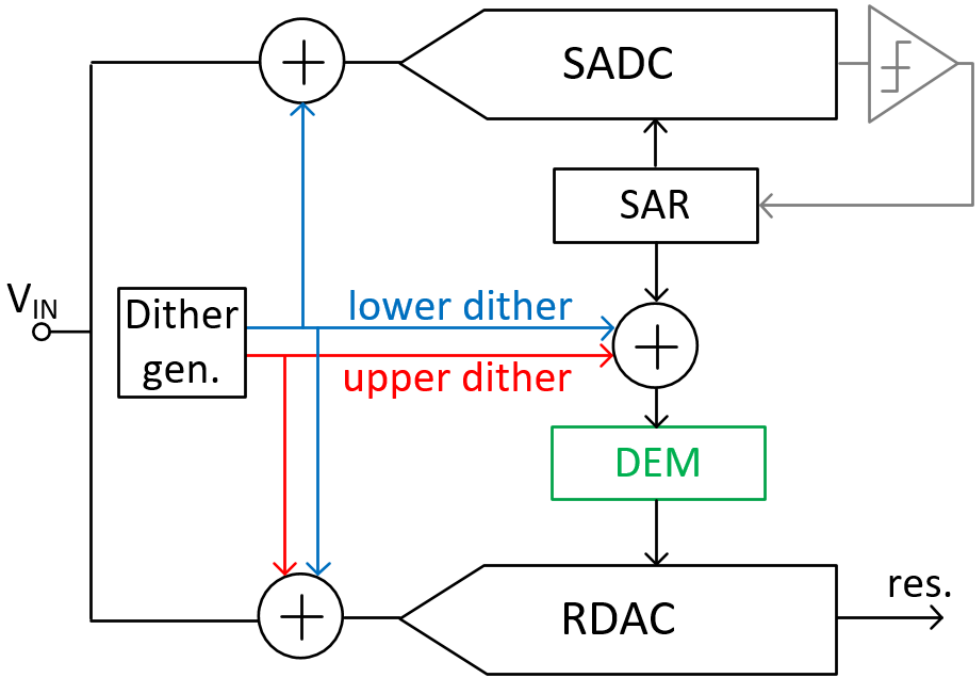
RDAC segmentation and DEM

Extra dither range

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Smaller DAC level shifters

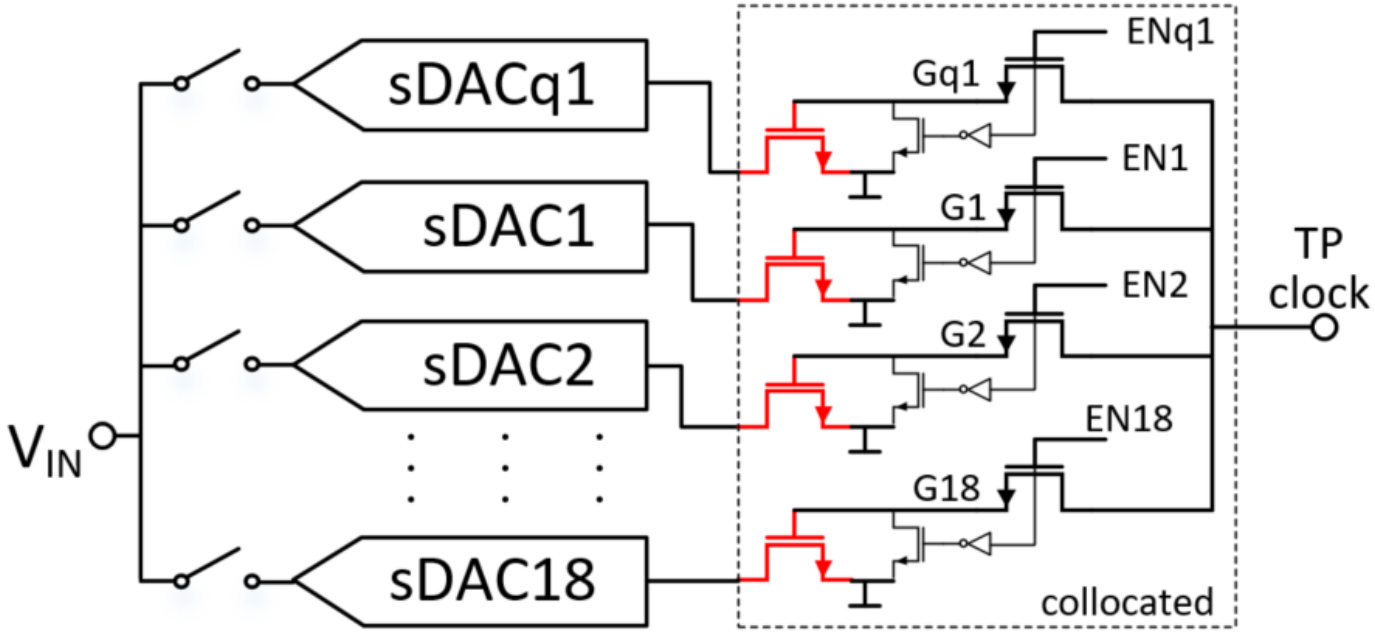
Power and area saving



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Managing Interleaving Errors

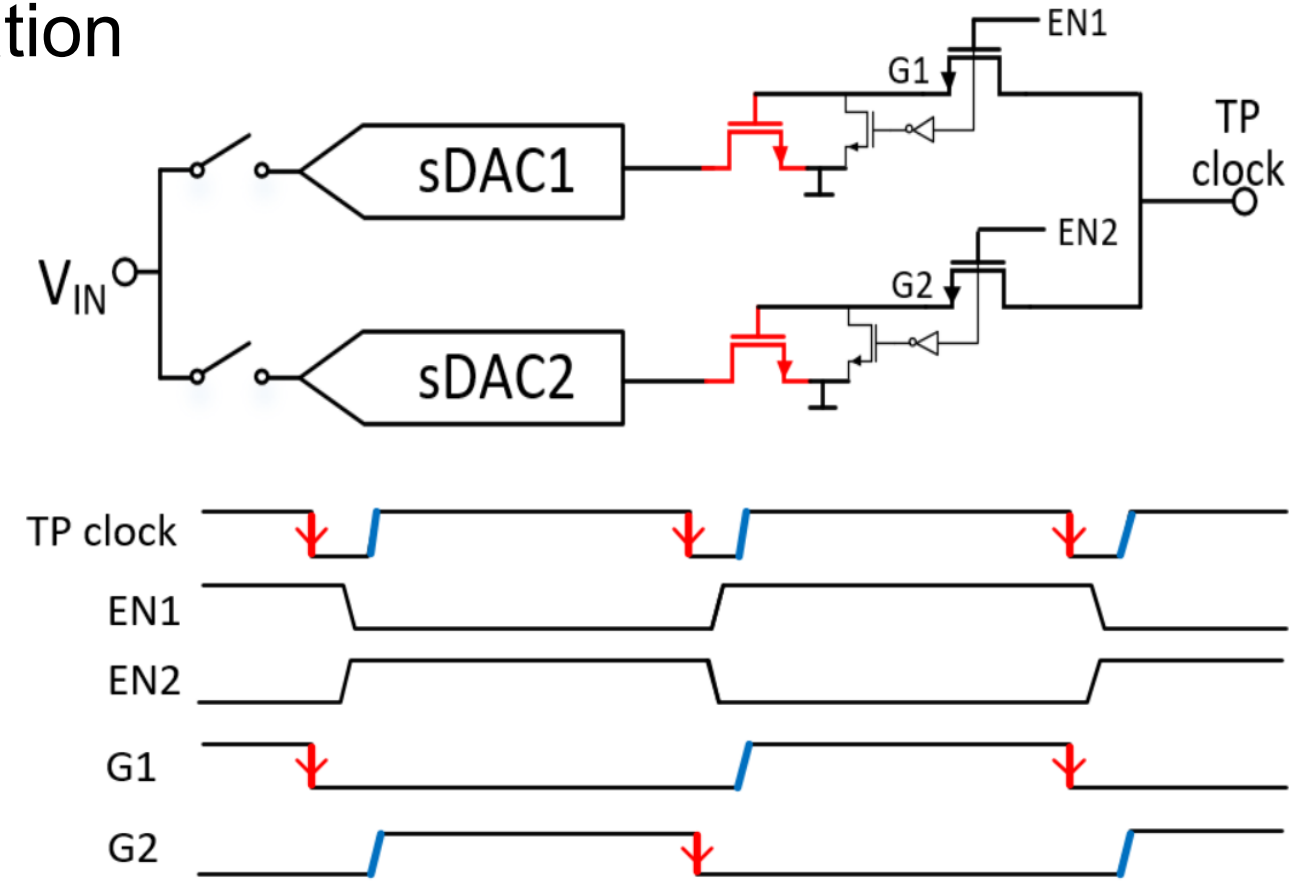
- Offset and gain – on-chip correction
- Bandwidth mismatch
  - DAC BW set to $15 \times f_{NYQ}$
- Timing mismatch
  - Matched clocking scheme



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Managing Interleaving Errors

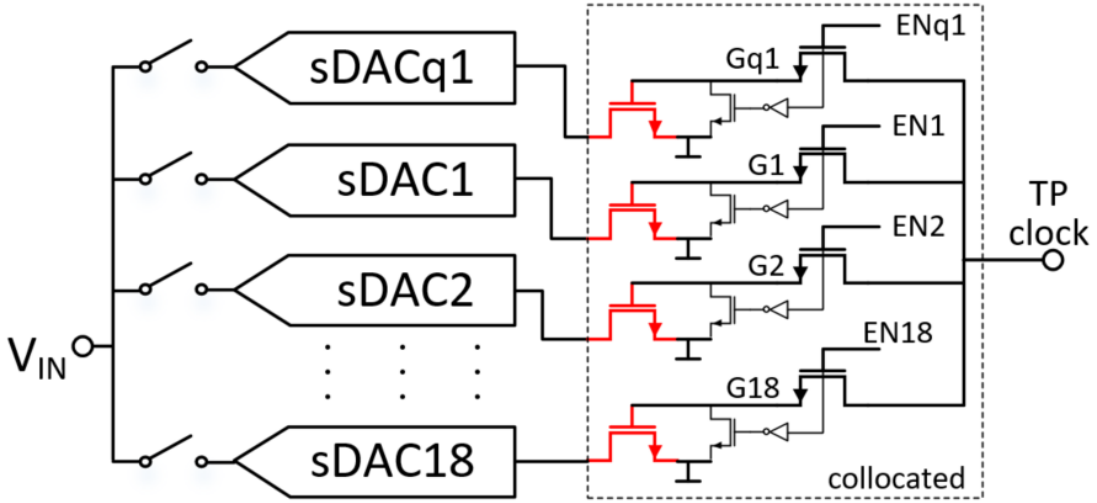
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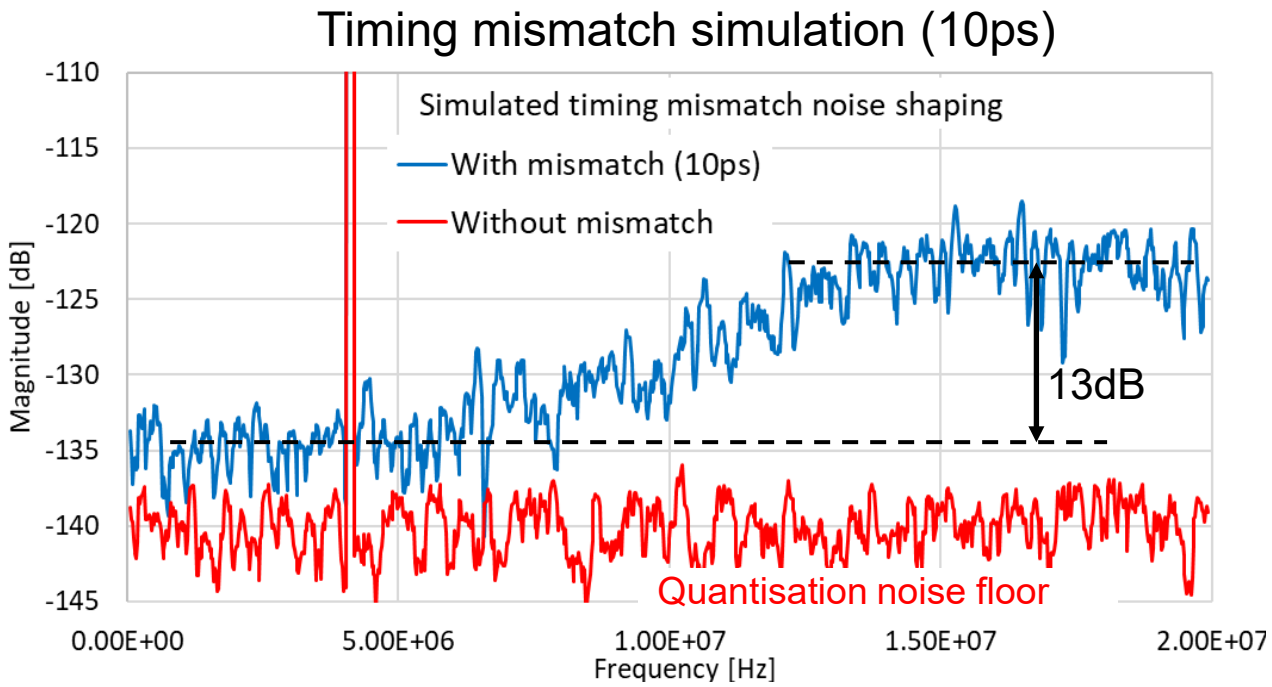


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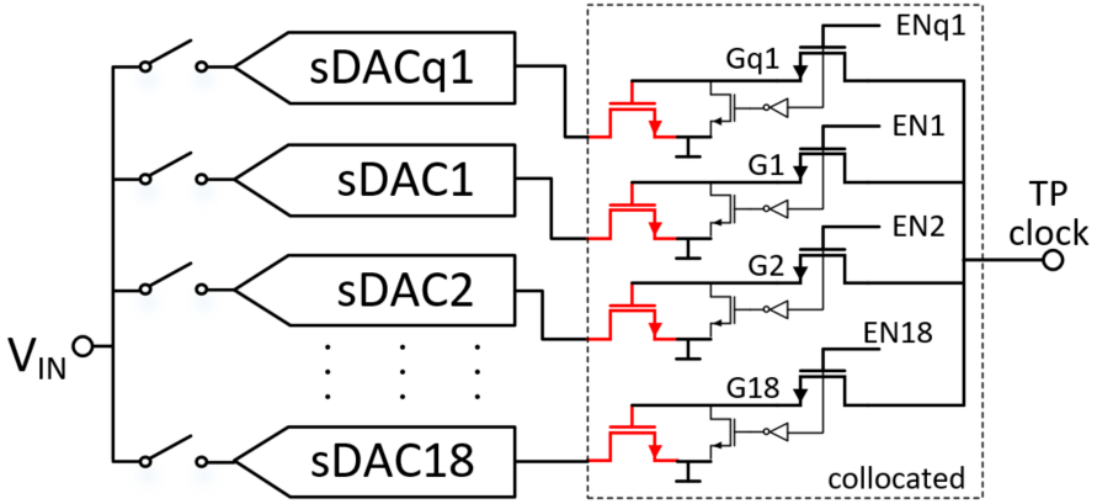


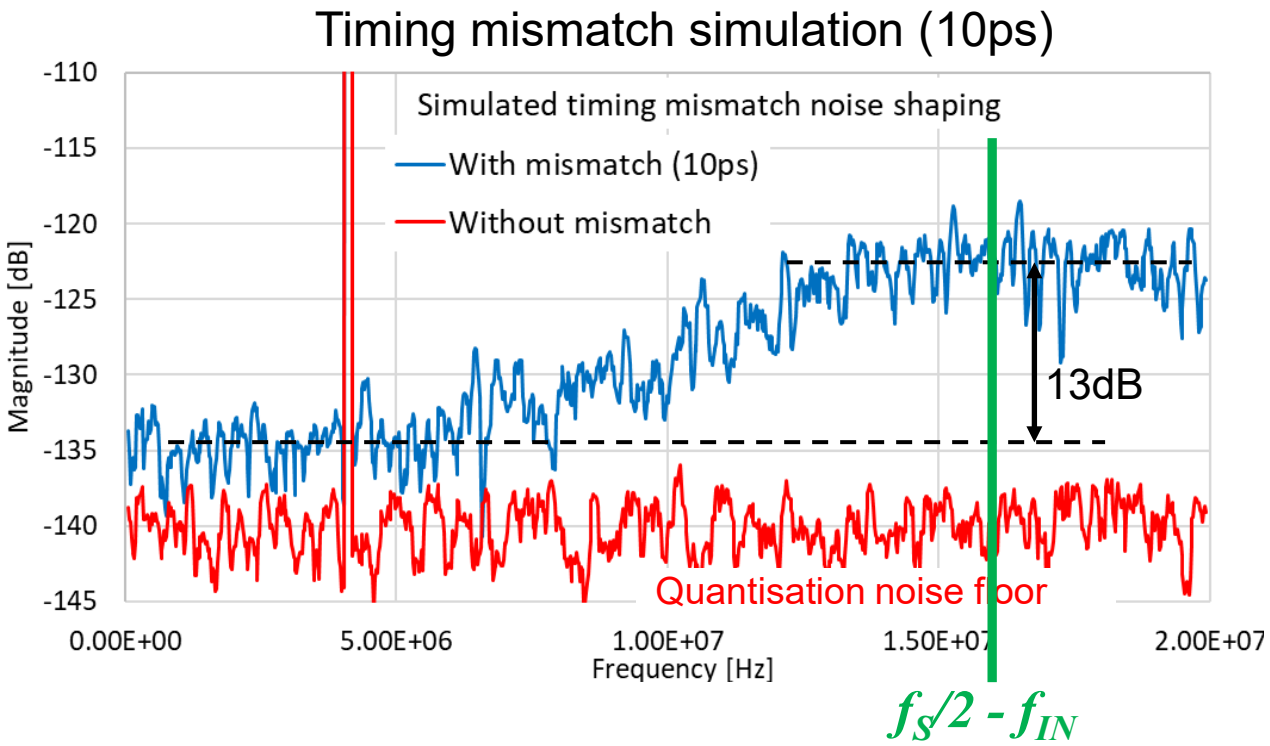


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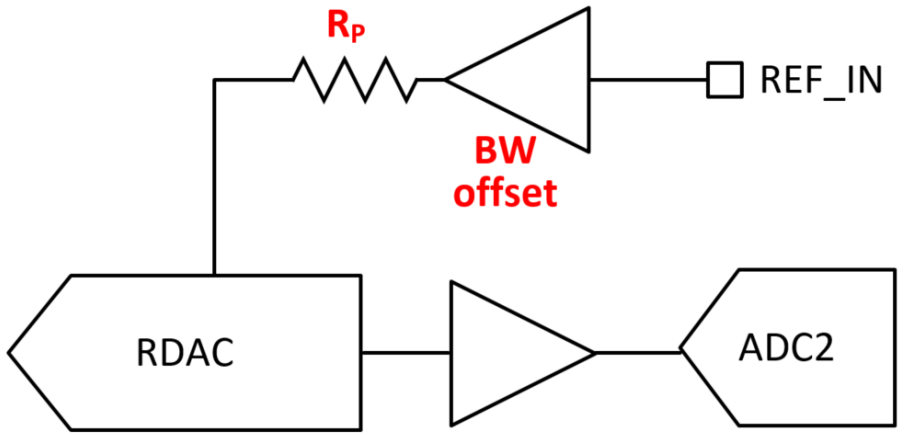


Interleaving tone with randomisation centred at $f_S/2 - f_{IN}$

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Reference Buffer

- Large RDAC for kT/C (20.5pF)
  - ~1.9mA current at 40MSps
  - CT reference buffer not suitable



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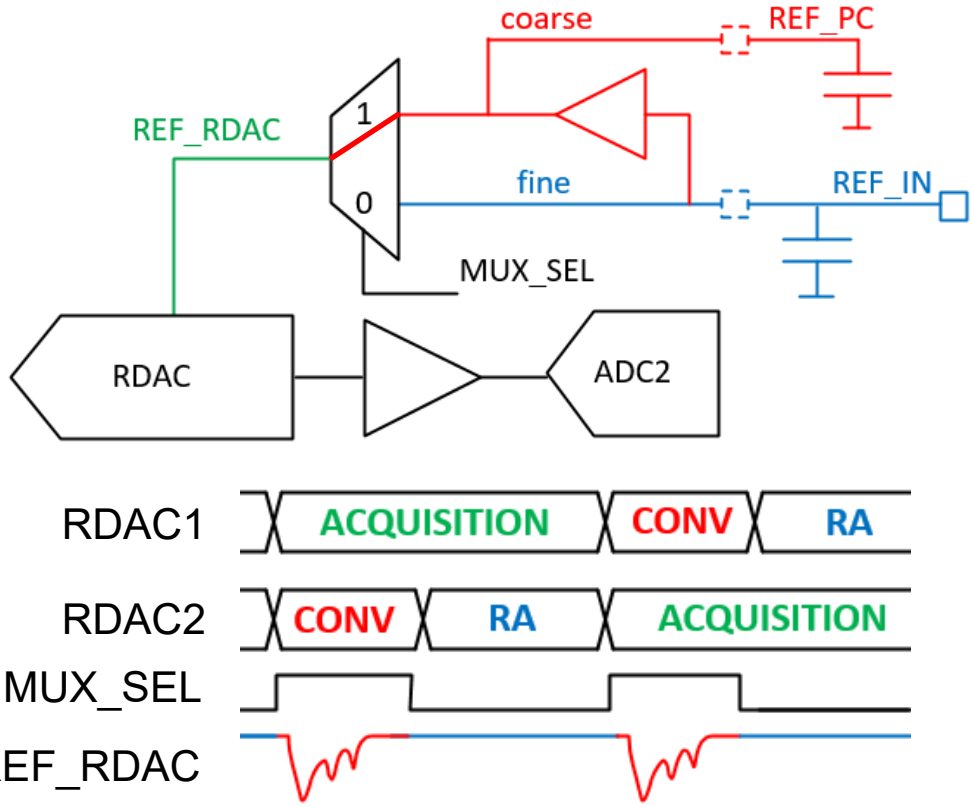
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Dynamic reference buffer

- Coarse signal during conversion



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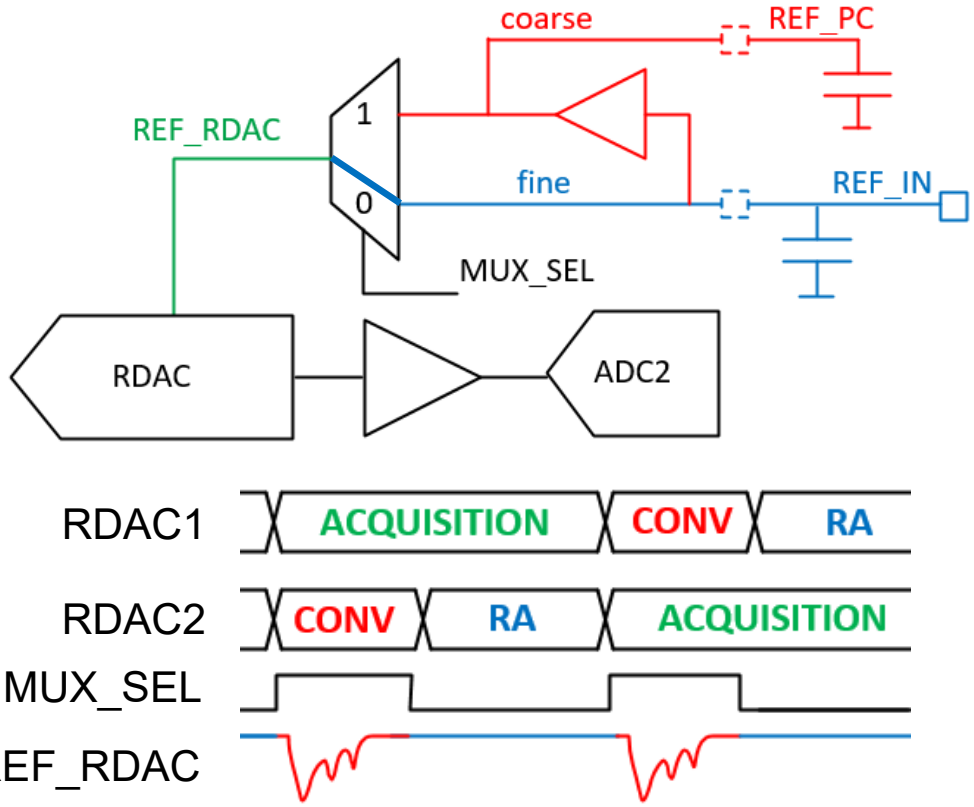
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- Fine (unbuffered) reference during RA



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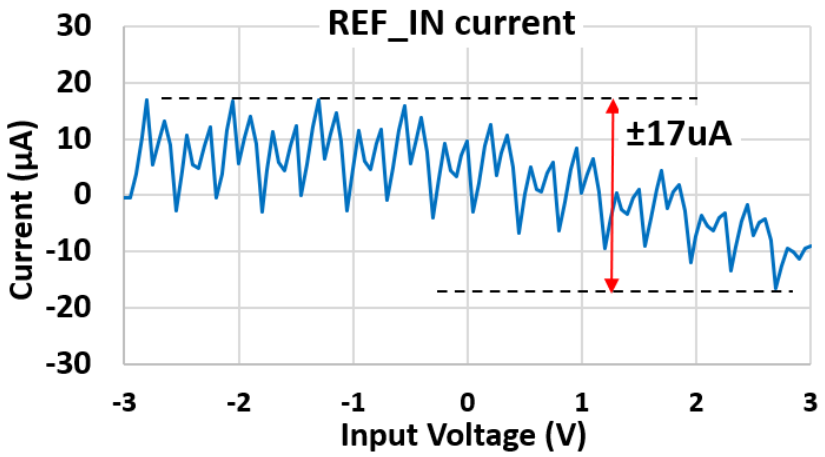
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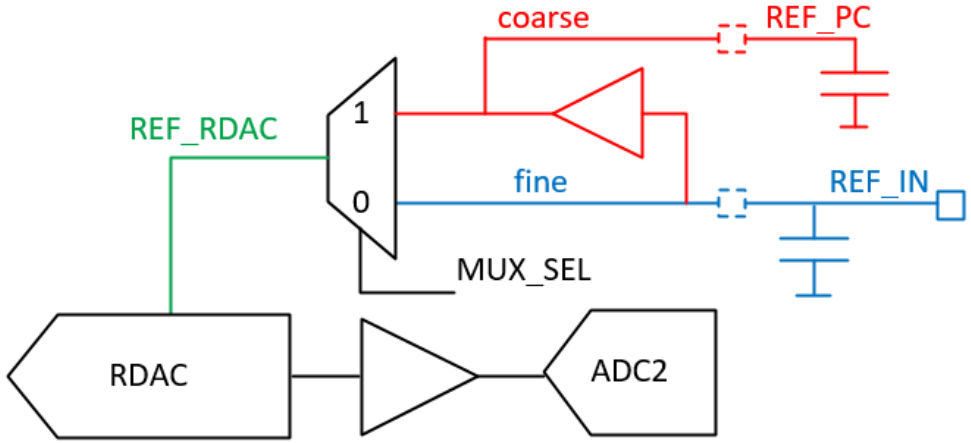
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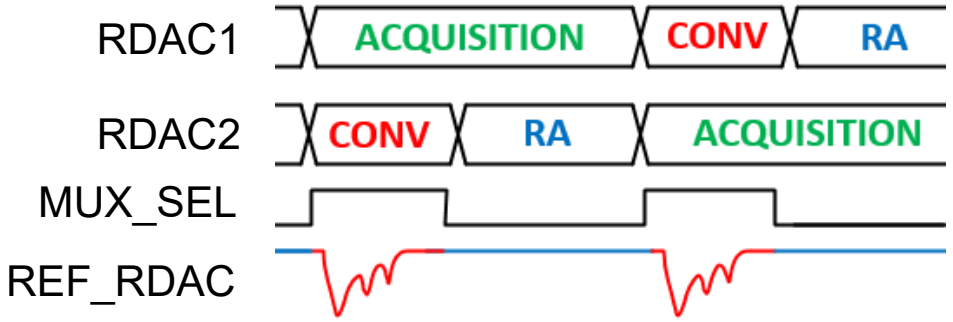
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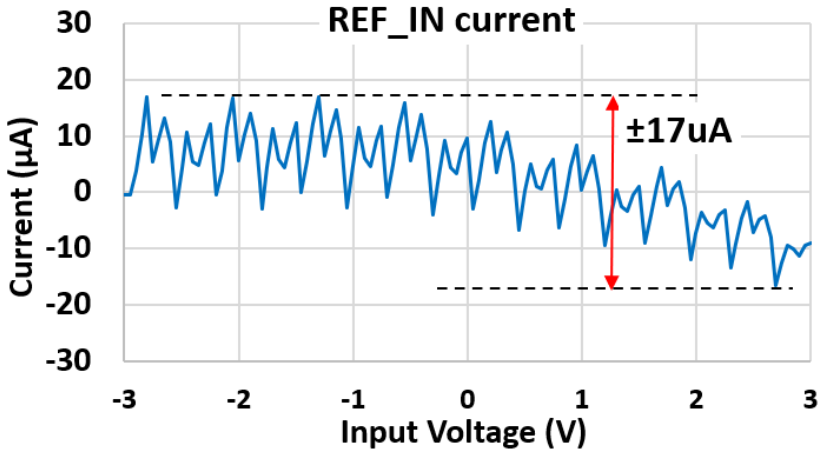
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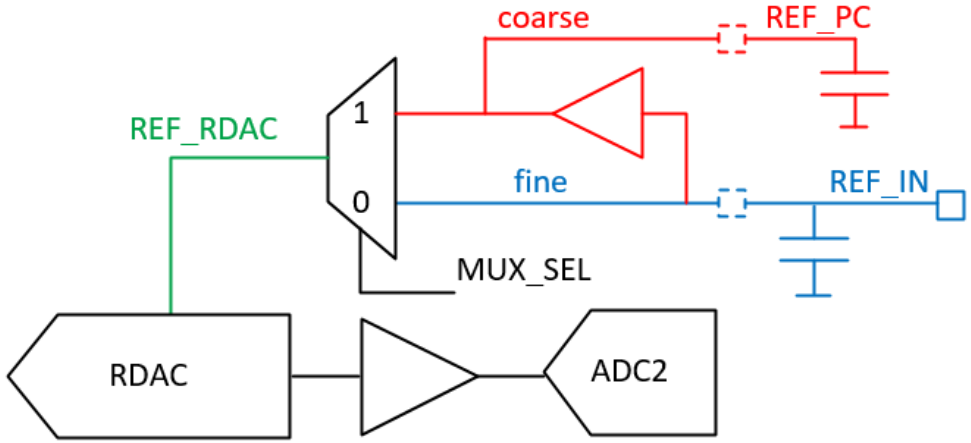
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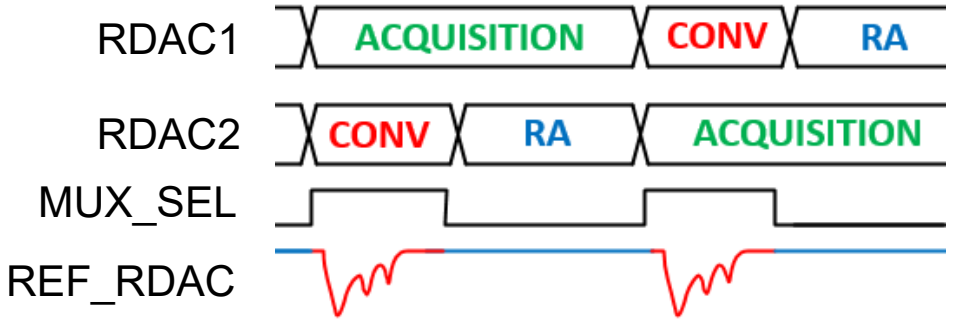
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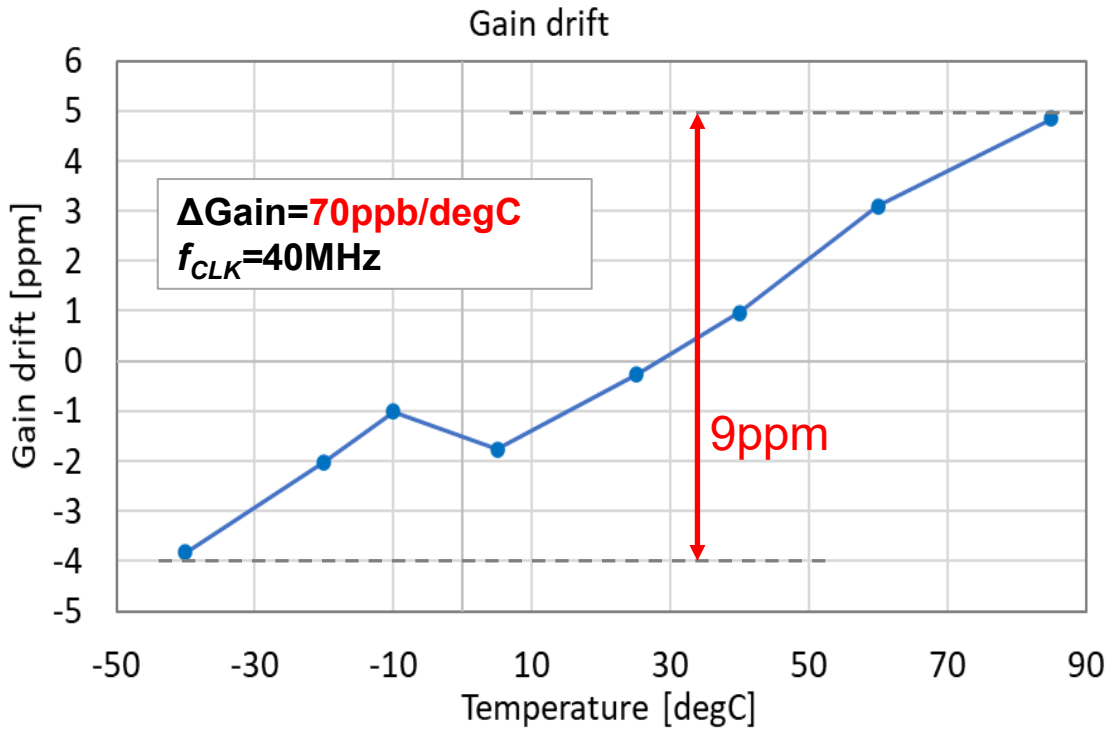


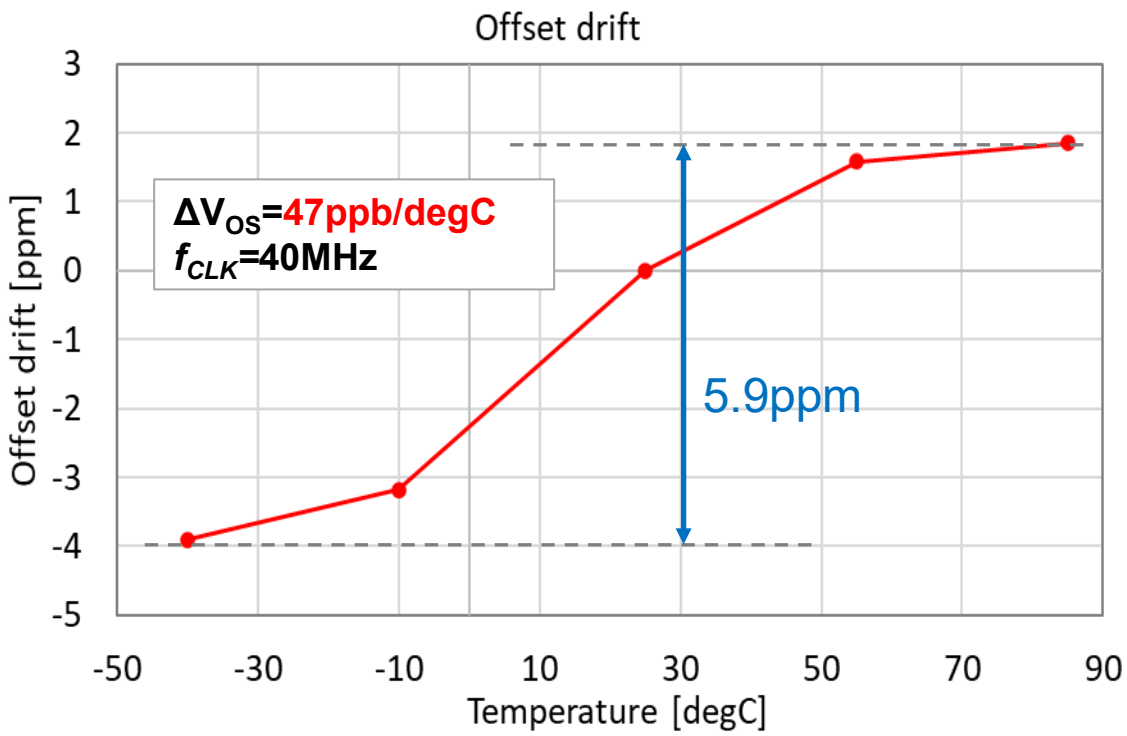


- **No offset and drift**
- **No 1/f noise**

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Average Gain and Offset Drift





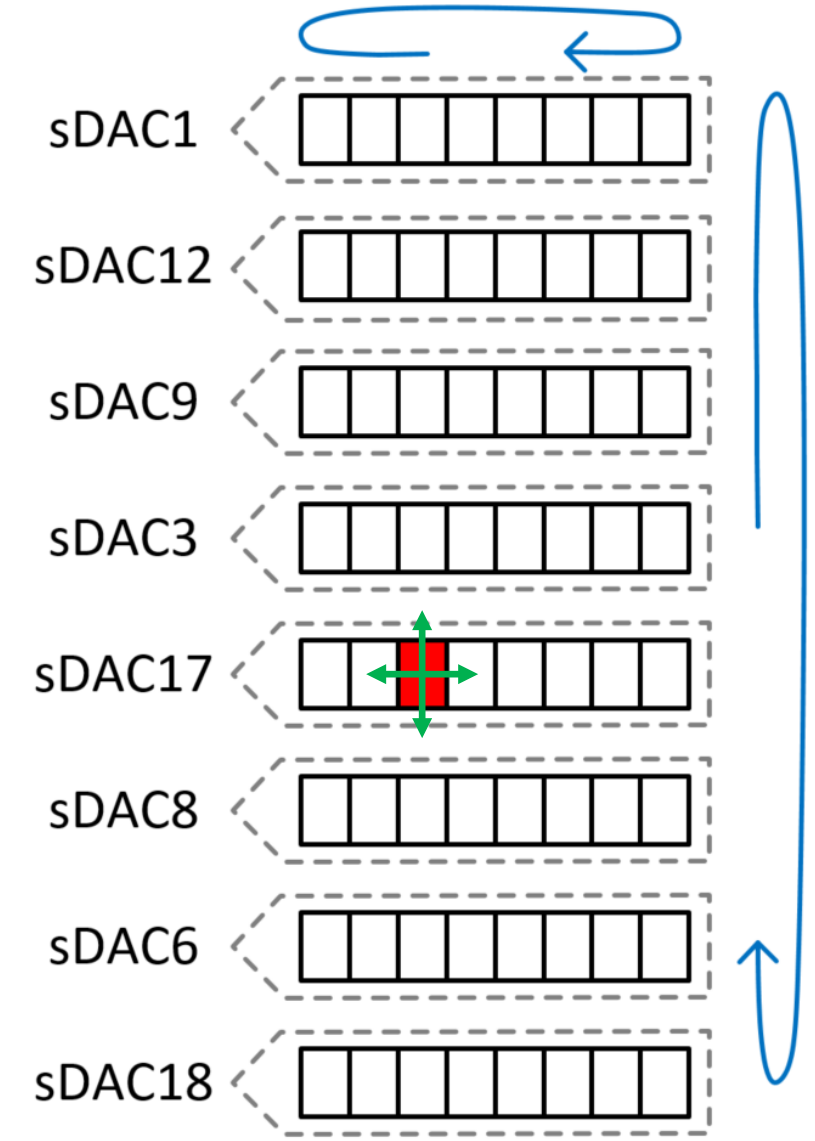
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DEM and Interleaving

DEM - limited by speed

3 dimensions of DEM

- Selection of the slices - 8/18
- 3b horizontal segmentation
  - Round robin shuffler
- 3b vertical segmentation



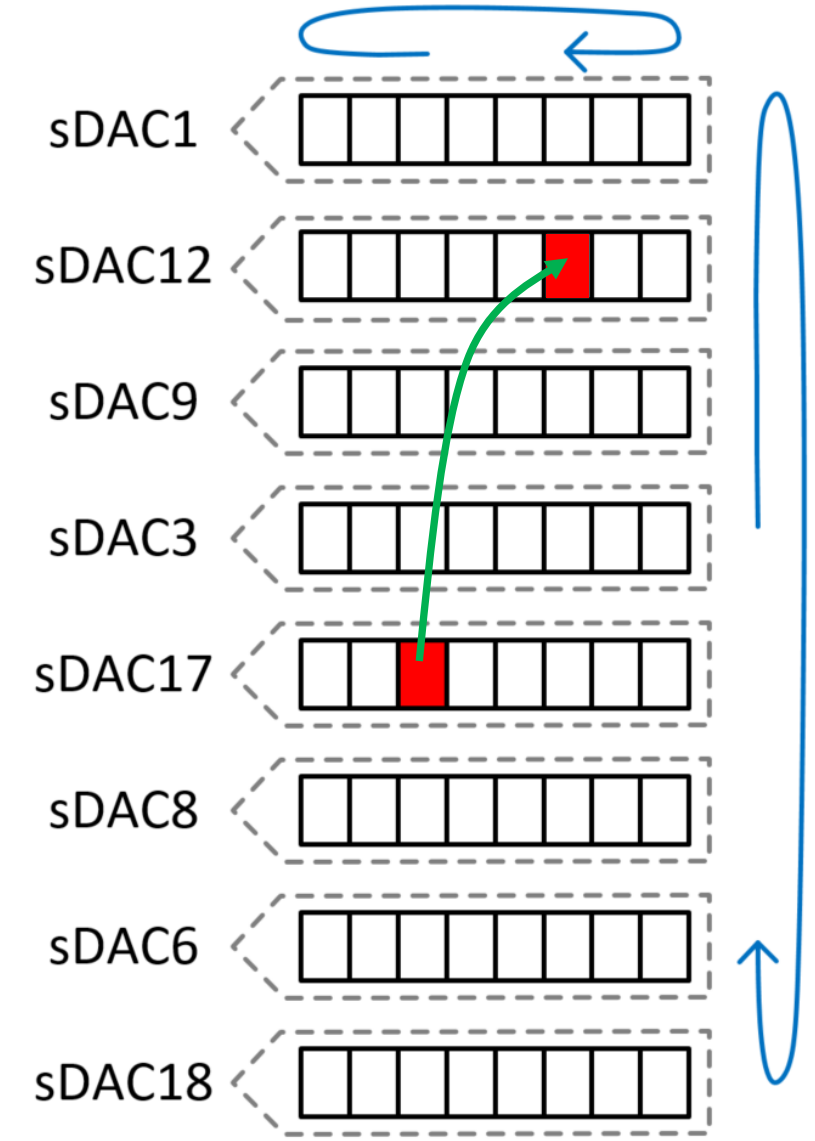
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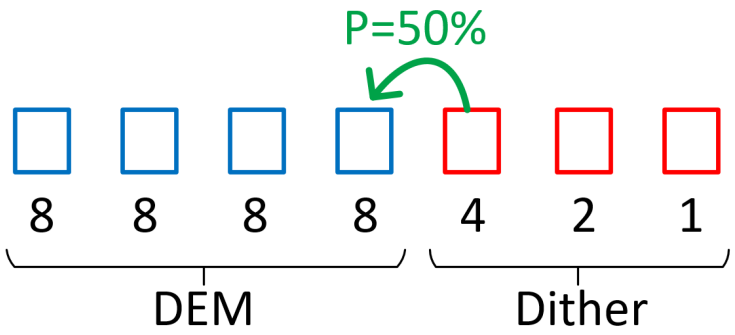
DEM and Interleaving

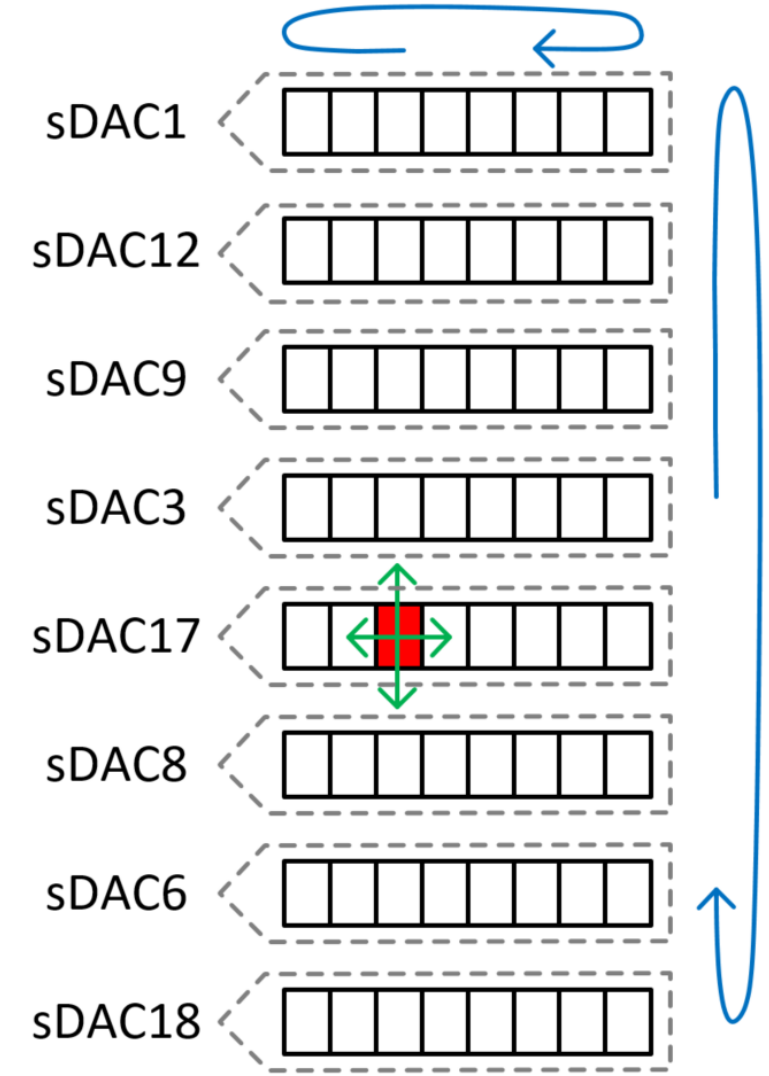
DEM - limited by speed

3 dimensions of DEM

- Selection of the slices - 8/18
- 3b horizontal segmentation
  - Round robin shuffler
- 3b vertical segmentation

Binary to unary bridging



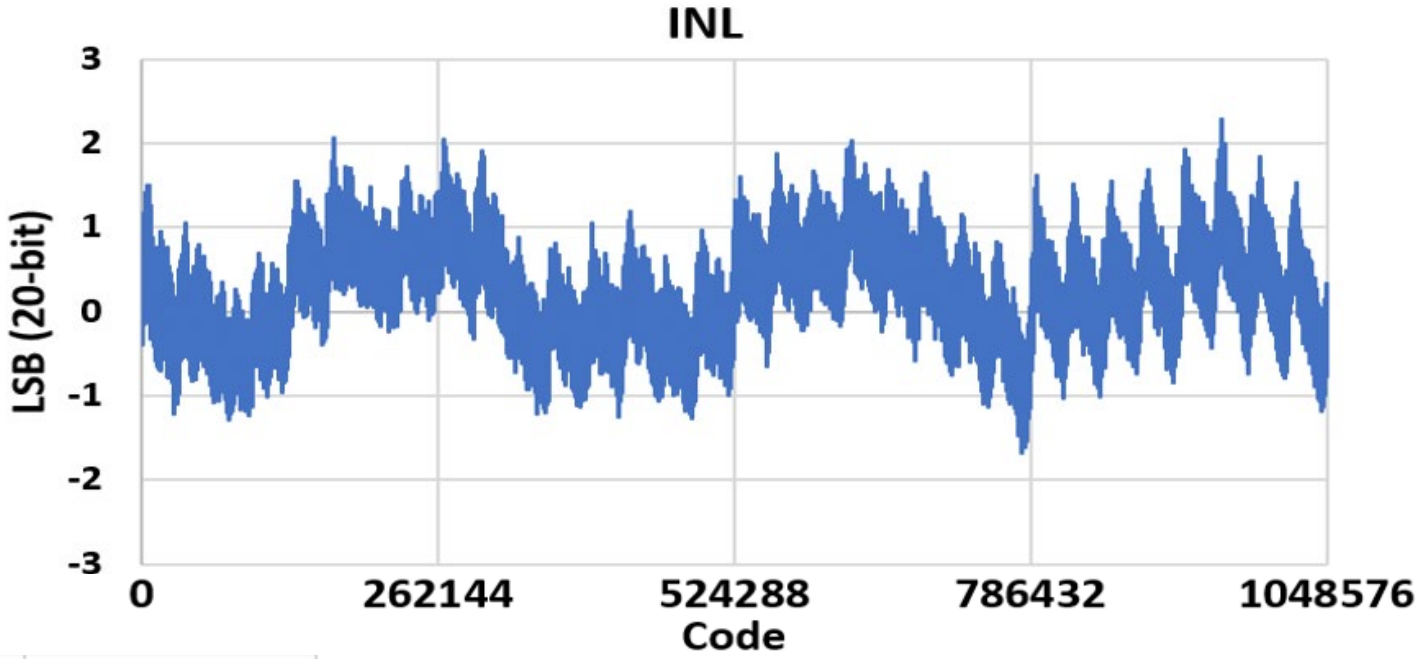


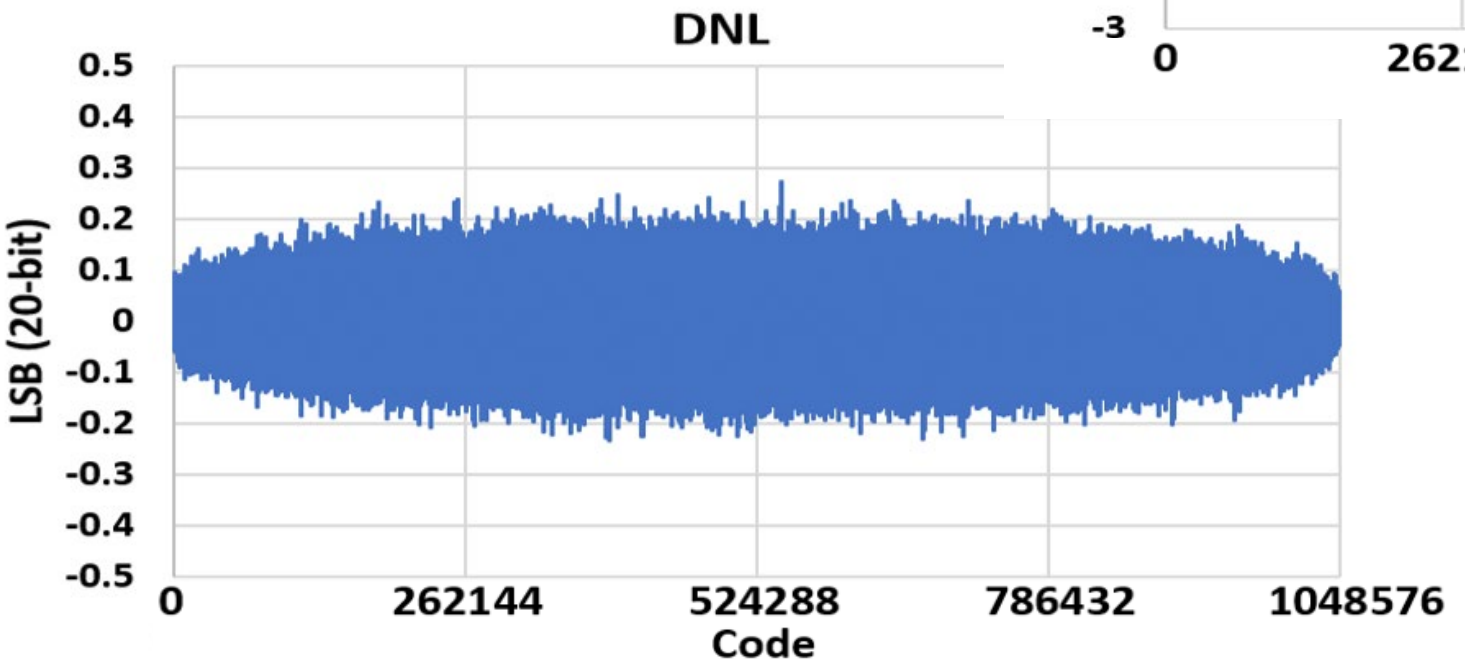
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INL and DNL

INL profile limited by:

input settling
digital crosstalk
reference settling

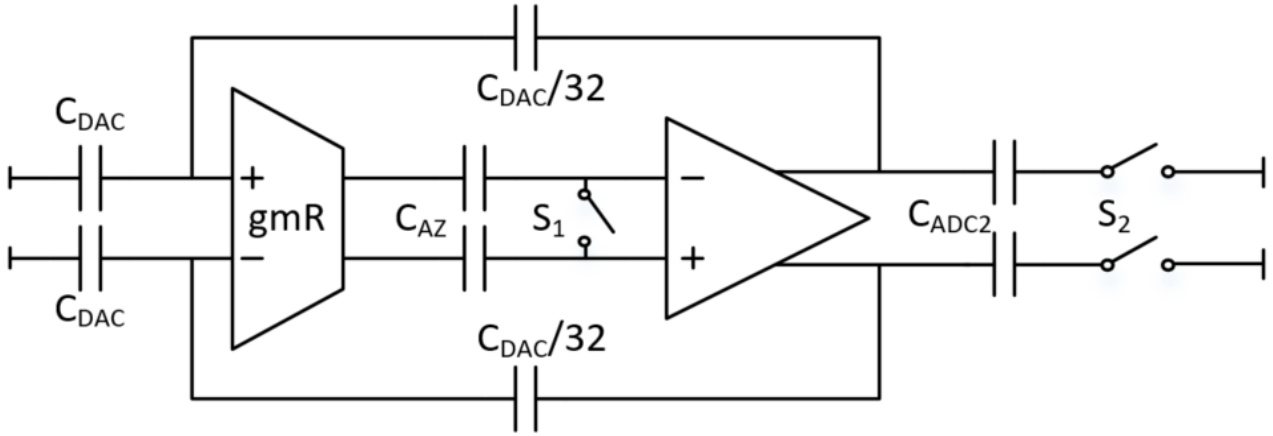




INL=2.2/-1.8LSB
DNL=±0.27LSB
Min hits/code ~250

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Residue Amplifier

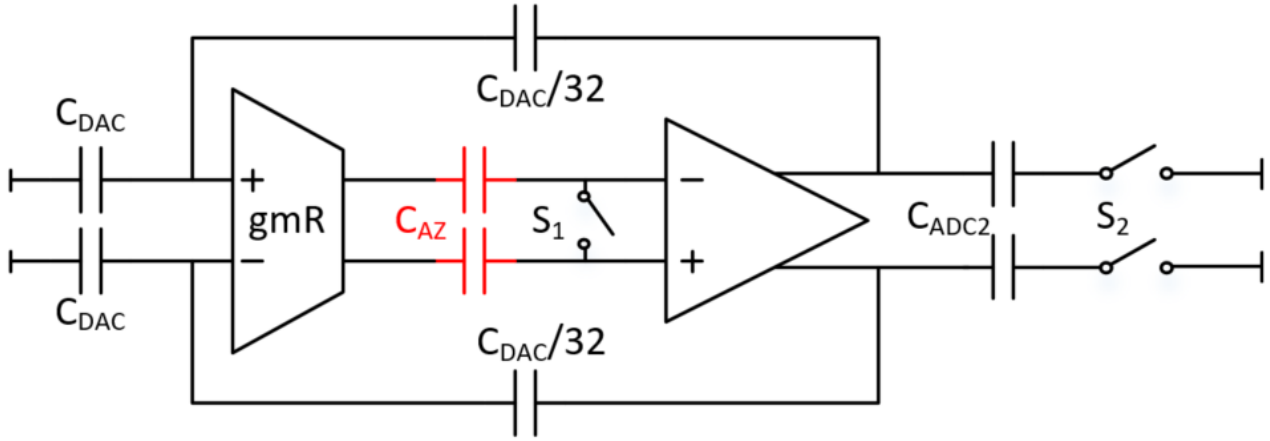


G_MR + OTA architecture

RDAC reference is still settling during RA phase

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Residue Amplifier



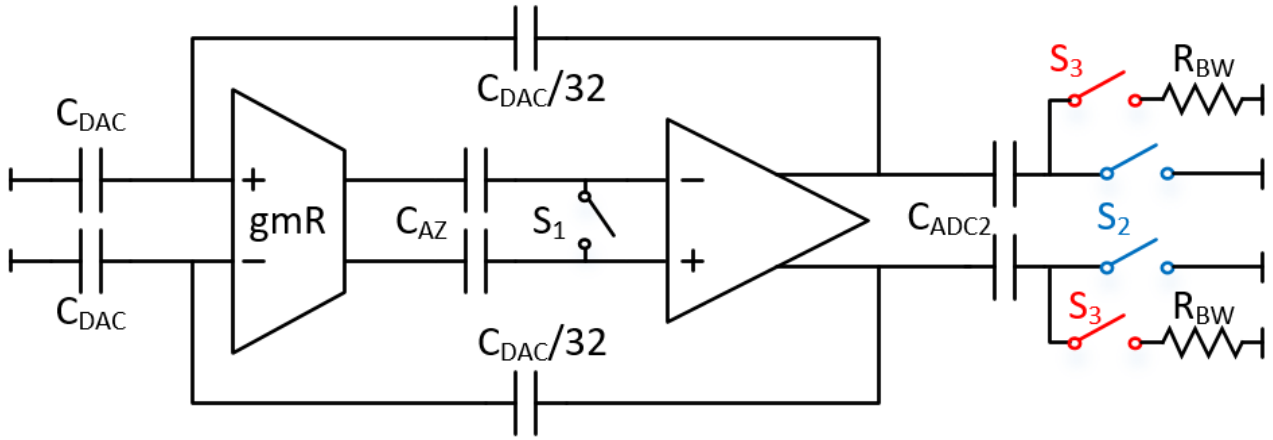
G_MR + OTA architecture

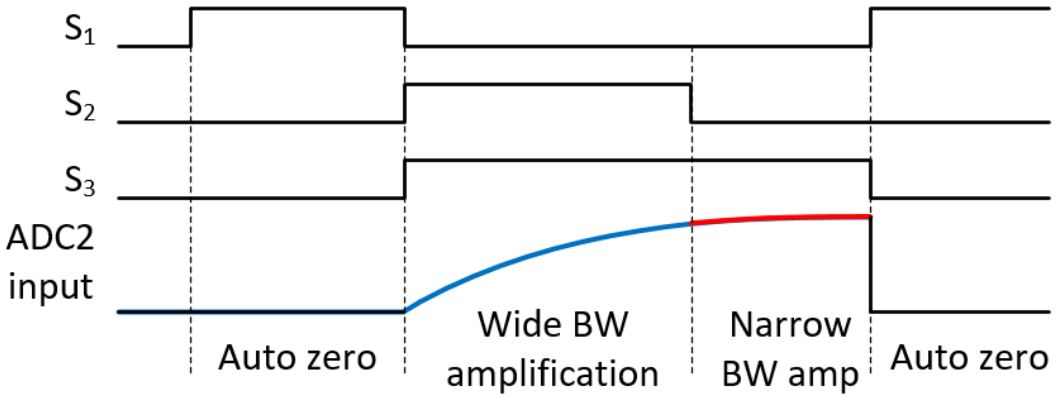
RDAC reference is still settling during RA phase

Auto-Zero removes offset and drift / noise penalty (-1.6dB)

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Residue Amplifier





G_MR + OTA architecture

RDAC reference is still settling during RA phase

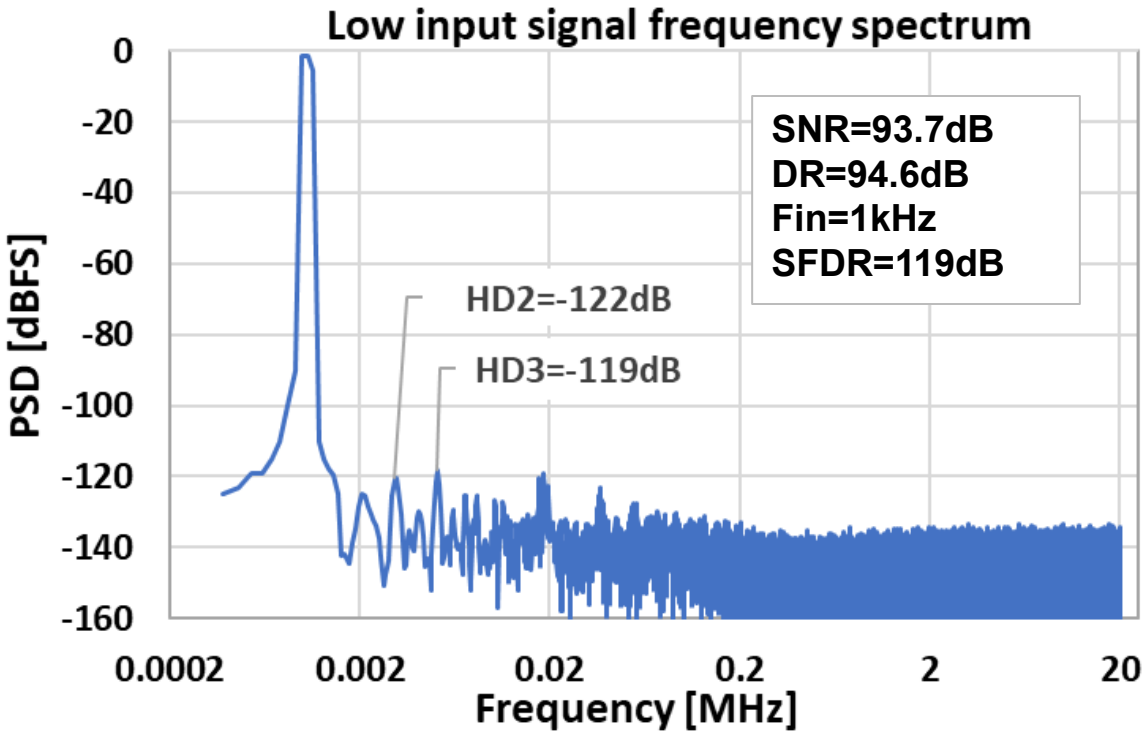
Auto-Zero removes offset and drift / noise penalty (-1.6dB)

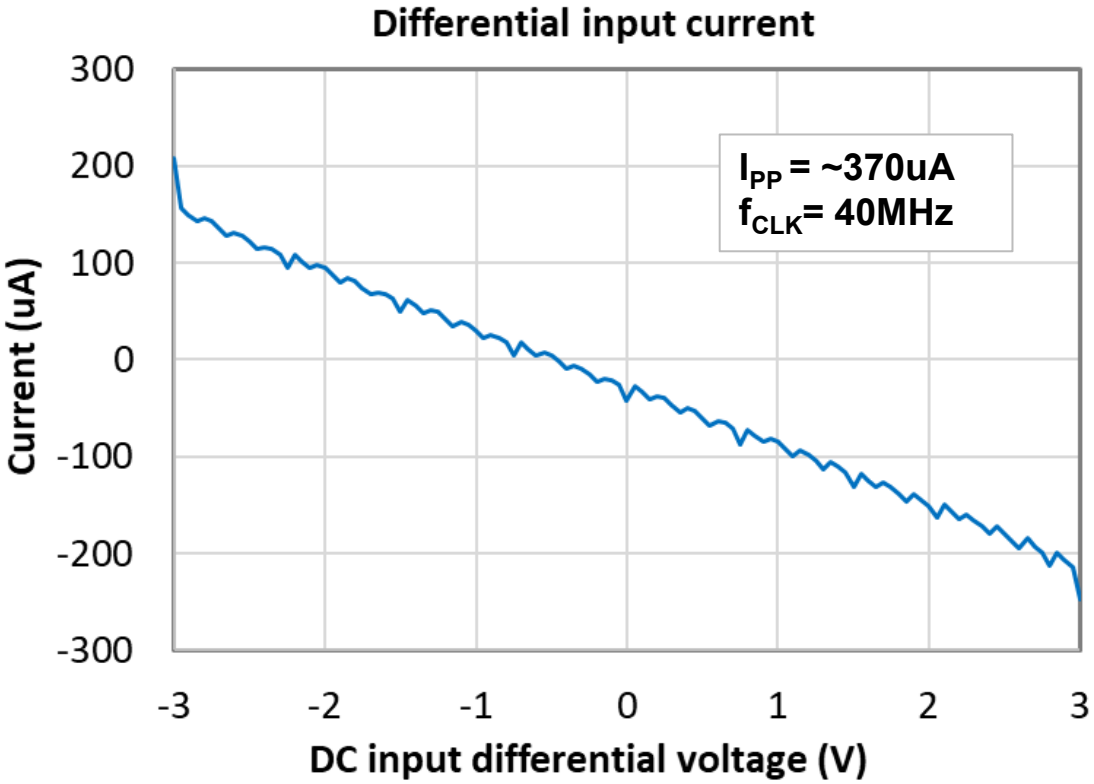
RA noise reduced by dynamic ADC2 sampling BW (+1.3dB)

Wide BW for settling / narrow BW for noise

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Low Frequency Performance (ADC + Driver)

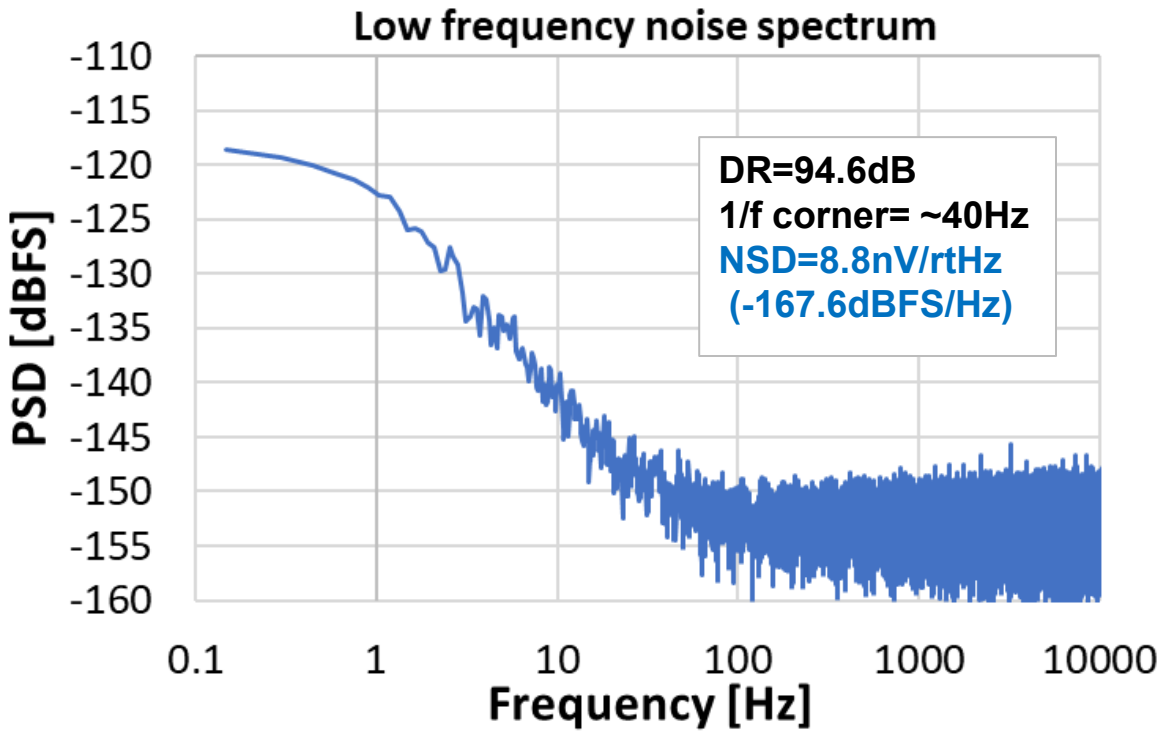


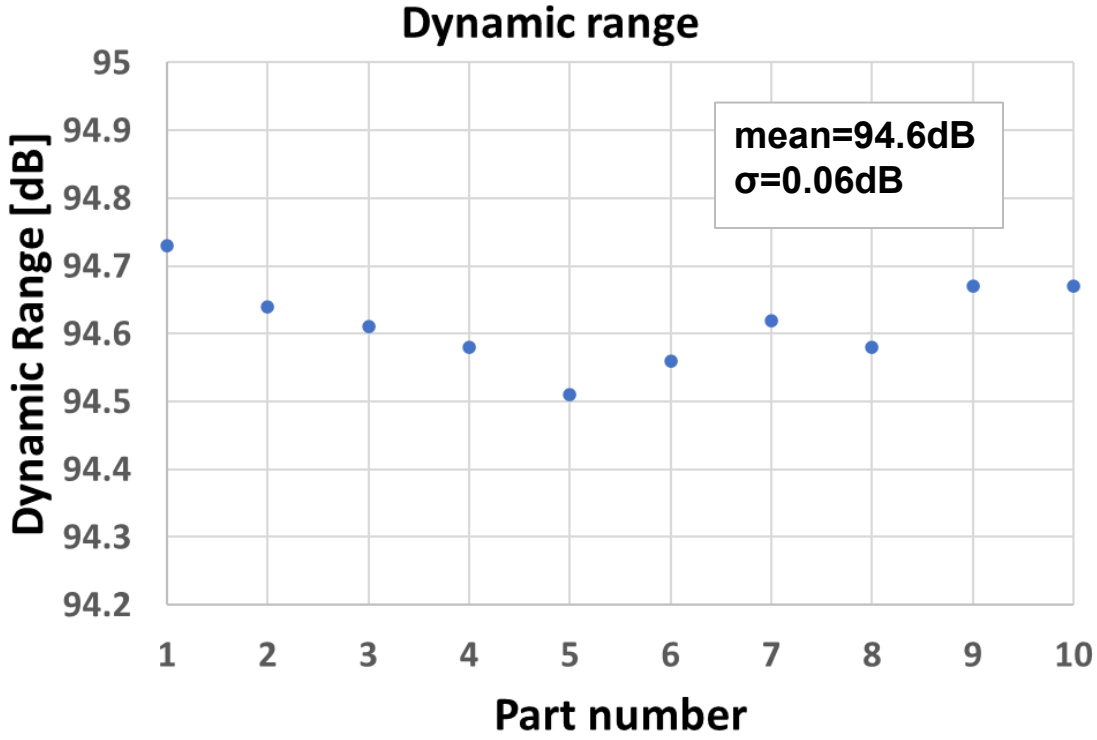


*SNR including AFE and source noise

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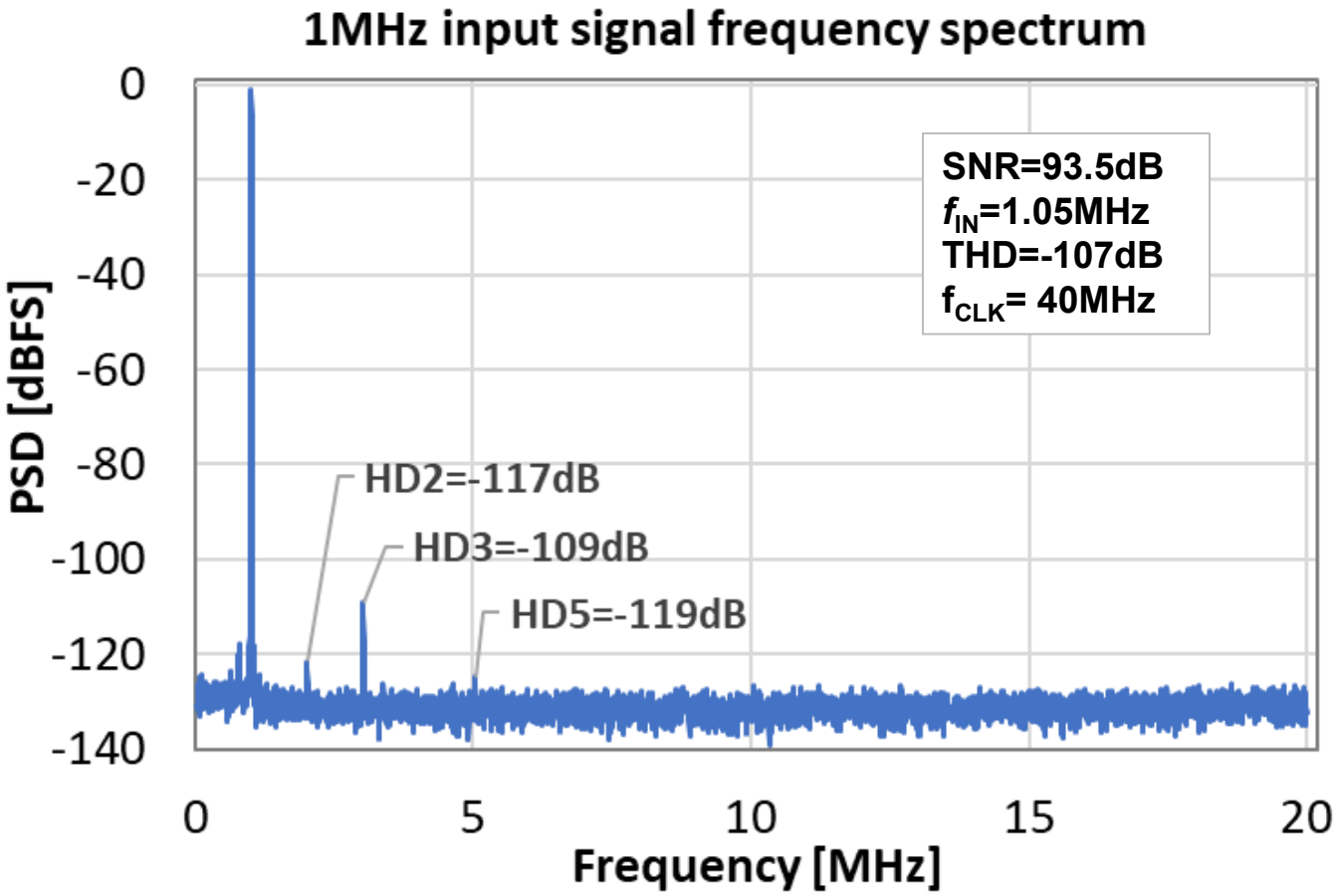
Low Frequency Performance

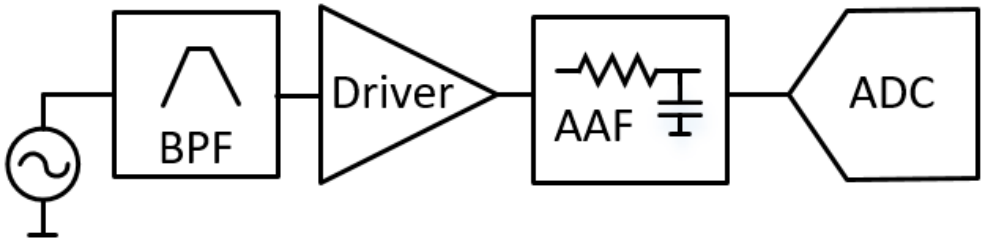




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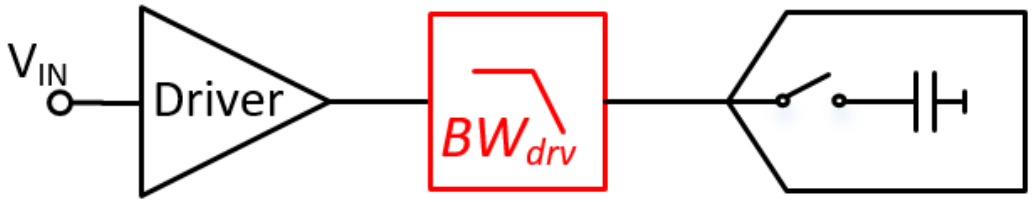
AC Performance



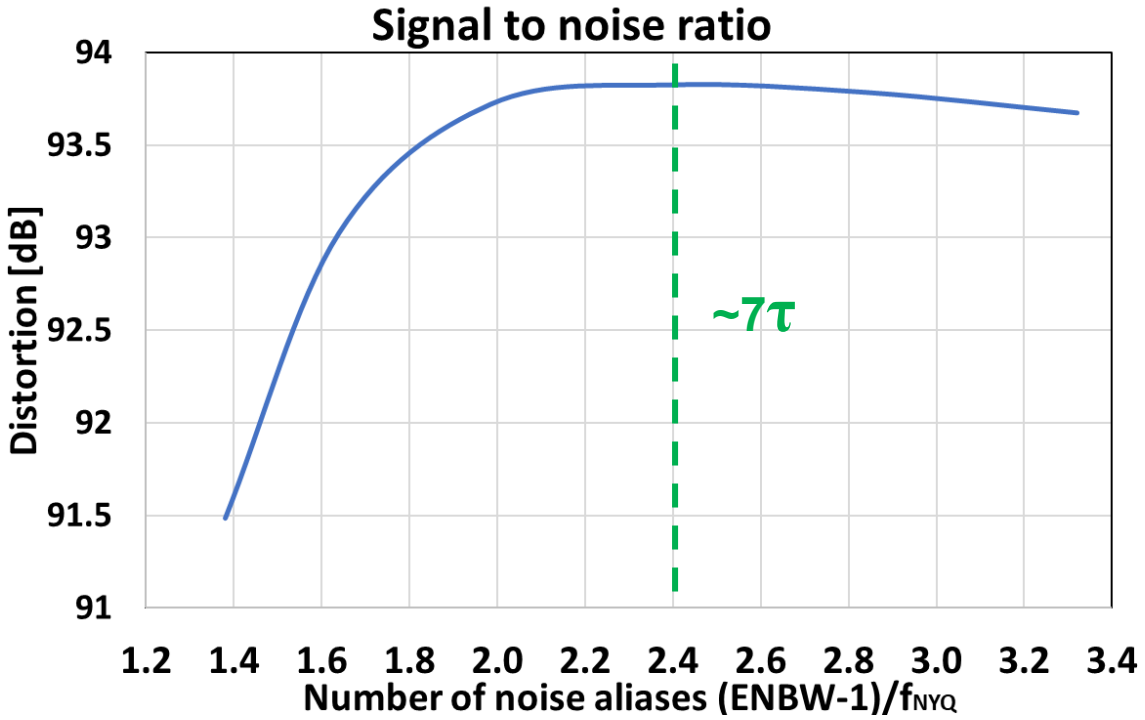


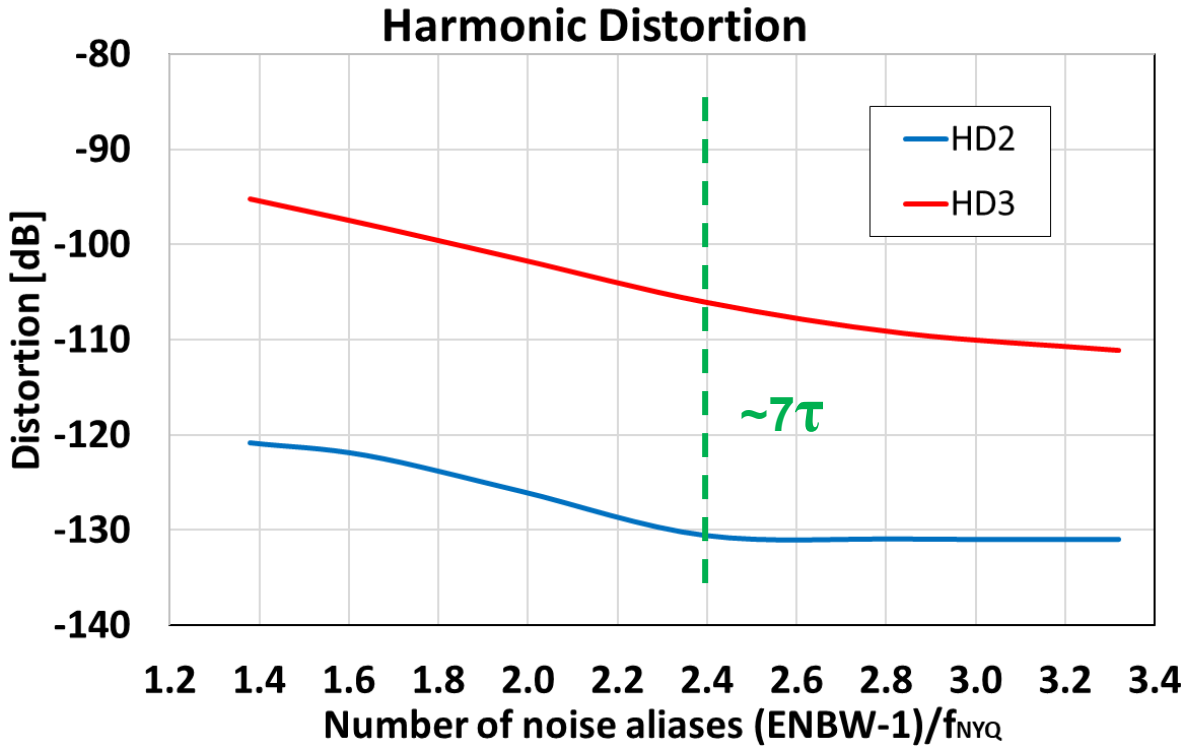
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Reducing the input filter bandwidth



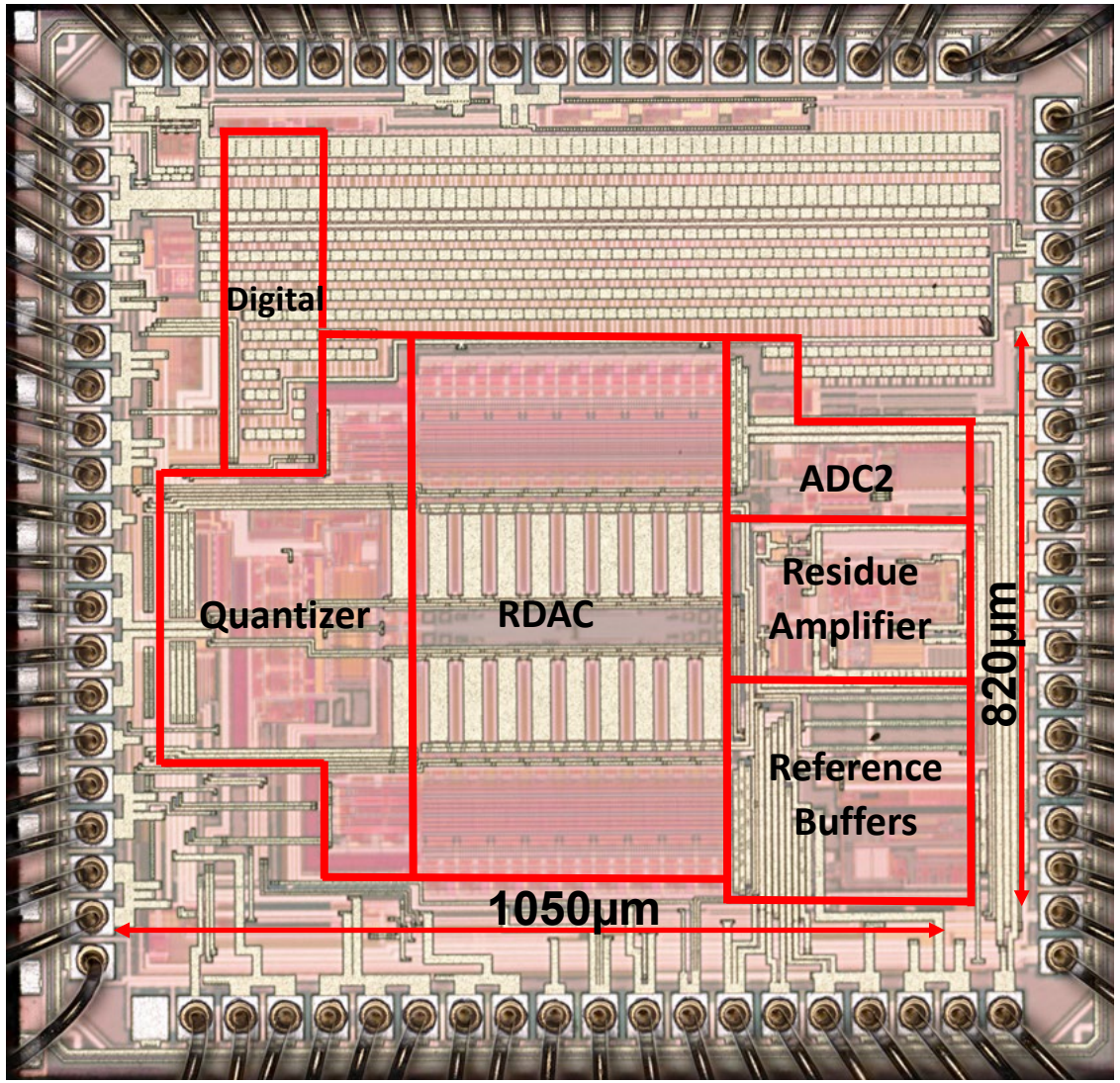
Continuous acquisition and 20b settling
Theoretical $ENBW_{DRV} = 7f_{NYQ}$





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Die Micrograph



Core area:
$A_{CORE}=0.77mm^2$

Core + ref. buffer area:
$A_{CORE+REFBUF}=0.9mm^2$

Process: 40nm ULP

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Comparison Table

	This work	Hurrell [1]	ElShater [2]	Li [3]	Bannon [4]	Steens-gaard [6]	Shen [8]	LTC2387 -18 [5]	ADC358 3 [7]
Process [nm]	40	250	180	180	180	180	55	-	-
F_S [MHz]	40	12.5	15	12	5	2	16	15	65
Resolution [bit]	20	18	16	16	18	24	16	18	18
Noise density [nV/rtHz]	8.8	26	17.2	18.8	21.9	17.1	103.9	15.9	11.8
INL [ppm]	±2.2	±32	±16	±24	±2	0.03	±36.8	±2.4	±28
SNR/DR [dB]	93.7/94.2	91/93	92.4/93.9	-/94	99/100.2	106/106.3	78/-	96/-	84.5/-
SFDR [dB]	119	93	100.7	100.8	-	-	98	119	95
Power core [mW]	50	105	9.8	9.43	30.5	8.5	16.3	-	-
Power system* [mW]	69	-	-	-	-	15	-	125	122
FOM_{S_DR} core [dB]	180.6	170.7	182.7	182.1	179.3	187.0	-	-	-
FOM_{S_DR} system* [dB]	180.2	-	-	-	-	184.5	-	-	-
FOM_{S_SNR} core [dB]	179.7	168.7	181.2	-	178.1	186.7	165	-	-
FOM_{S_SNR} system* [dB]	178.3	-	-	-	-	184.2	-	173.8	168.8

Power ADC core
- ADC1, ADC2
- Residue Amp.
- Comparators
- Timers
- Logic calibration

Power system
- ADC core
- Reference buffers
- Interface (LVDS)
- Protection circuitry
- CM buffer

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Conclusions

- **Two stage, 20b, 40MSps, SAR ADC with analog interleaving**
  - NSD=8.8nV/rtHz / -167.6dBFS/Hz
  - SNR=93.7dB, DR=94.6dB, FOM=180.6/180.2dB
    - 1/f corner: 40Hz
  - INL=+2.2/-1.8ppm, DNL=0.27ppm
    - Offset drift=47ppb/degC, Gain drift=0.07ppm/degC
  - Power: ADC core=50mW, full-chip=69mW
- **Signal-chain friendly**
  - Continuous signal acquisition
    - 3-4x relaxation of the anti-aliasing filter bandwidth
    - Compatible with low-power drivers
  - ~4dB extra improvement on signal chain FOM

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- **Management:** Damien McCartney, John Quill, Ray Goggin, Joe Creech, Christian Birk

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Thank you for your attention.

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